“Just One More Clip”: Short Videos, Big Self-Control Problems*

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Abstract

I study self-control problems in media consumption and how short-form design amplifies them. Short units repeatedly reactivate temptation that lasts longer than each unit of content, turning local temptation into sustained overconsumption. Using microdata from a U.S. short-drama platform, I exploit a nonlinear top-up menu to infer ex-ante viewing plans and show that paying users watch 82.1% more than intended. Structural estimates imply that temptation lasts 11.2 minutes on average, short relative to the full drama but long relative to one-minute episodes. Counterfactual default time limits and mandatory breaks improve consumer welfare. An extrapolation to TikTok highlights the broader welfare relevance of short-form design.

JEL codes: D12, D91, I31, L82

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1 Introduction

Digital platforms increasingly monetize attention by making consumption granular, continuous, and easy to extend. These design choices have become central to both policy debates and legal challenges. In *K.G.M. v. Meta et al.*, a recent bellwether case in Los Angeles Superior Court, a jury found Meta and Google negligent in the design of products alleged to induce compulsive use.[1] The case does not, by itself, establish an economic mechanism. It does, however, sharpen a central question for economic analysis: when do platform-design choices generate welfare losses rather than merely satisfy high demand?

This paper studies that question in the context of short-form media consumption. Self-control problems arise when consumers' ex-post choices depart from their ex-ante preferences, which have been documented in settings such as gym attendance, retirement saving, and digital addiction.[2] Short-form media is a particularly important setting because platforms can shape the decision environment directly: they choose how finely consumption is divided, how easily users can continue, and when stopping decisions arise. Policy discussions now range from default time limits to outright social media bans, requiring systematic evaluation of the inefficiencies caused by self-control problems.[3] This paper contributes by using a structural approach to quantify the welfare losses and evaluate policy counterfactuals.

The paper's core mechanism is that content length interacts with the *duration* of temptation. Users derive welfare-relevant utility from media consumption, but they also experience short-run temptation for a finite period. When each unit of content is shorter than this temptation horizon, the next unit remains tempting when the current decision ends. Temptation therefore rolls forward with consumption, turning "just one more clip" into sustained overconsumption. Longer content attenuates this mechanism because users evaluate later portions of the content outside their current temptation horizon. Although I develop the argument for short-form media, the insight is more general: breaking a tempting good into smaller units can increase consumption by making it easier for consumers to repeatedly give in to local temptation, thereby exacerbating self-control problems.

I formalize this mechanism in a stylized model of sequential short-video consumption. Naïve users do not fully anticipate that temptation will shift forward as they continue watching. They therefore plan to stop after a small number of videos but repeatedly revise that plan when the next decision point arrives. The model shows that shorter content expands the set of users who experience self-control problems and increases the associated surplus loss. It also yields a policy implication: a default time limit can improve welfare for users whose temptation duration is

[1] The verdict remains subject to appeal. TikTok and Snap, which were also defendants, settled before trial. See Reuters (2026); Huamani and Ortutay (2026); Crowell & Moring LLP (2026).

[2] See, for example, Laibson, Repetto, Tobacman, Hall, Gale, and Akerlof (1998); DellaVigna and Malmendier (2006); Kan (2007); Allcott, Gentzkow, and Song (2022).

[3] For instance, in 2023, TikTok introduced a feature that sets a default 60-minute daily screen time limit for all accounts held by users under the age of 18 (Source: TikTok NewsRoom). In a more extreme move, on November 21, 2024, Australia introduced a bill in parliament to ban social media use—including TikTok—for children under 16 (Reuters, November 20, 2024, "Australia launches 'landmark' bill to ban social media for children under 16").

shorter than the limit, because these users do not expect ex ante to watch beyond it.

Empirically, I study this mechanism in the short-drama industry, a fast-growing form of mobile entertainment that combines serialized drama with short-form video design. Short dramas are typically divided into dozens of minute-long episodes. The platform in my data serves the U.S. market by acquiring or producing these dramas, promoting them through major social media platforms, and monetizing consumption through episode unlocking. This setting is well suited for studying self-control problems because consumption is sequential, each decision unit is short, and continuation decisions are tightly linked to platform design.

Using drama-level aggregate data, I first document that dramas with shorter average episode length have higher completion rates. Across 353 drama series on the platform, after controlling for language, genre, release month, episode count, first-episode viewership, and video bitrate, an additional minute of episode length predicts a 2.42-percentage-point lower completion rate. This effect is equivalent to 21.2% of the mean completion rate. The evidence is descriptive, since episode length is chosen jointly with content, production, and marketing decisions. It nevertheless shows that the content-length margin emphasized by the model is visible in platform-wide data.

I then focus on user-level behavior for a superstar drama on the same platform. The drama consists of 80 one-minute episodes. The first nine episodes are free; subsequent episodes must be unlocked sequentially using non-refundable platform tokens. Users buy tokens through a nonlinear top-up menu consisting of four packages at different price points, where larger packages offer a lower price per episode. This pricing structure turns package choice into a revealed-preference measure of planned continuation. A user who buys a larger package reveals a willingness to commit to more future episodes at a lower unit price, while a user who repeatedly buys the smallest package reveals limited planned continuation at each purchase but continued viewing ex post.

The reduced-form evidence shows substantial time inconsistency. Among users who purchase at least one package, the first top-up choice implies an intended consumption of 28 minutes, but users ultimately watch 51 minutes on average. Actual consumption therefore exceeds intended consumption by 82.1%. In addition, users who repeatedly buy small packages often spend more than they would have spent by purchasing a larger package upfront.[4] Among paying users, 39.1% spend more than the ex-post rational benchmark, with an average overpayment of $5.51, or 22.7% of optimal expenditure. The data also reveal repeated purchases of small packages that are difficult to reconcile with Bayesian learning. Under learning, repeated experience should reduce uncertainty and sort users toward upgrading after favorable signals or stopping after unfavorable ones. Instead, users become increasingly likely to purchase the smallest package again, pointing to a persistent local-repurchase force rather than experimentation alone.

A key contribution of the paper is to identify the distribution of temptation duration through a structural model in which heterogeneous users make dynamic discrete choices over top-up packages and episode unlocking. The user's state includes token balance, habit stock, and posterior

[4] Users in my sample spend between $29.99 and $69.96 to watch the full 80-minute drama, suggesting substantial overpayment. For comparison, an adult ticket for a movie at the Princeton Garden Theatre cost $13.50 in 2024. The much higher willingness to pay for this short drama highlights the strength of demand in this setting.

belief about drama value. The model incorporates three distinct mechanisms: temptation affects perceived short-run utility, learning affects beliefs about drama quality, and habit formation raises the marginal utility of later episodes within a viewing round. The nonlinear top-up menu is central for identification. Repeated small-package purchases identify the magnitude of temptation. The shares of different package sizes identify the distribution of temptation durations, because users with longer temptation horizons anticipate more continuation and are more likely to choose larger packages. Transitions after early small-package purchases help separate learning from temptation: learning predicts sorting toward upgrading or stopping, whereas temptation predicts persistent repurchase of small packages. Habit formation is identified from state dependence in continuation and top-up choices within viewing rounds.

The estimates imply that short-lived temptation is the main driver of overconsumption. The estimated mean per-episode drama value is -2.2¢, with a standard deviation 10.3¢. By contrast, temptation is large: the estimated temptation magnitude is 23.3¢ per minute, more than ten times the absolute value of the mean episode value. Temptation is also heterogeneous in duration. An estimated 28.8% of users have no temptation, while the remaining users draw positive temptation durations from a heavy-tailed distribution. The average duration is 11.2 minutes, with a standard deviation of 22.6 minutes. This duration is short relative to the full 80-minute drama but long relative to one-minute episodes, so temptation remains active across many consecutive episode-level decisions and can be repeatedly re-triggered by the short format. Learning, by contrast, is rapid. Posterior uncertainty about drama quality falls by 61% after the first episode and has fallen by 87%, to 1.3¢, by the time users make their first top-up choice. I also find evidence for habit formation: each additional episode watched within a round raises subsequent per-episode utility by 1.6¢ on average. These estimates indicate that persistent overconsumption is driven primarily by short-lived but repeatedly renewed temptation.

The estimated model implies large welfare losses from self-control problems, amplified by short content length. Average user surplus is $-\$0.25$ in the baseline model. Eliminating temptation raises average surplus to \$0.58, implying a temptation-induced loss of \$0.83 per user. These losses are largest for users near the margin of valuing the drama, for whom temptation induces additional consumption that is not justified by welfare-relevant utility. Extending episode length from one to eight minutes lowers the average temptation-induced surplus loss from \$0.83 to \$0.73, recovering 12.4% of the one-minute benchmark loss. Longer episodes reduce the payment margin most closely associated with repeated temptation-driven continuation, while leaving completion approximately stable among users with sufficiently high continuation value.

I evaluate three counterfactual policies: package-size requirements, default time limits, and mandatory breaks at the top-up stage. All policies increase user surplus by targeting the interaction between short content and temptation duration. Removing the \$4.99 package raises average user surplus by \$0.46. A default time limit tied to the first top-up choice also improves consumer surplus: with an opt-out cost of \$0.50, user surplus rises by \$0.31. Mandatory breaks generate smaller but positive surplus gains, raising user surplus by \$0.18 with a five-minute break.

Finally, I extrapolate the estimated preference parameters to broader short-video platforms such as TikTok with a back-of-the-envelope exercise. In the one-hour short-video benchmark, average hourly surplus is $-\$0.44$. Eliminating temptation raises hourly surplus to \$1.72, implying a temptation-induced surplus loss of \$2.16 per hour. Increasing video length to 12 minutes could recover \$1.29 per hour, or 59.6% of the one-minute temptation loss. The optimal default time limit is 10 minutes, which recovers \$1.28 per hour, or 59.1% of the loss. Aggregating using 150 million U.S. monthly active TikTok users implies a monthly temptation-induced welfare loss of \$11.7 billion, whereas a default time limit would recover \$6.9 billion per month. These calculations rely on strong extrapolation assumptions, but they illustrate the scale of the welfare stakes when short-form design interacts with self-control problems.

This paper makes four contributions. First, it introduces the duration of temptation as a key object in the analysis of self-control problems and applies this framework to explain why short-form content is particularly addictive. Second, by estimating the distribution of temptation durations, the paper provides field-based evidence on the understudied question of how long temptation lasts. Third, it develops an approach to separately identify self-control problems and habit formation, showing that both mechanisms influence user behavior but have distinct welfare implications. Finally, the paper quantifies the welfare losses caused by short-form content in monetary terms and evaluates policy interventions that can mitigate these losses.

Related literature. Building on the behavioral framework of present bias and self-control problems, my model extends these theories to explain addictive behavior in short-form content consumption (e.g., Laibson, 1997; O'Donoghue and Rabin, 1999; Gruber and Köszegi, 2001; Gul and Pesendorfer, 2001; DellaVigna and Malmendier, 2004; Gul and Pesendorfer, 2007). Theoretically, the paper contributes by showing how content length interacts with temptation duration, highlighting the role of short-format design in exacerbating impulsive consumption.[5] Empirically, the paper provides further evidence of self-control problems in a digital platform setting, complementing evidence from health-club memberships (e.g., DellaVigna and Malmendier, 2006), cigarette consumption (e.g., Kan, 2007; Ho, 2025), and retirement savings decisions (e.g., Laibson, Repetto, Tobacman, Hall, Gale, and Akerlof, 1998; Laibson, Chanwook Lee, Maxted, Repetto, and Tobacman, 2024).[6] Interpreting temptation as present bias, the paper also contributes to the understudied question of how to define the duration of the present by estimating its distribution in the field, whereas existing evidence comes largely from laboratory experiments (Augenblick, 2018; Balakrishnan, Haushofer, and Jakiela, 2020).

My work also complements the literature on digital addiction. It builds on Allcott, Gentzkow, and Song (2022), who use a field experiment to estimate a model incorporating habit formation, self-control problems, and naïveté. Other studies on digital addiction primarily focus on its consequences (e.g., Vanman, Baker, and Tobin, 2018; Allcott, Braghieri, Eichmeyer, and Gentzkow, 2020; Mosquera, Odunowo, McNamara, Guo, and Petrie, 2020; Braghieri, Levy, and Makarin,

[5]This insight connects to the concept of "skewness preference in the small," as proposed by Ebert and Strack (2015) in the context of prospect theory and naïveté.

[6]The presence of monetary payments makes it possible to identify time-inconsistent preferences in my setting. See Strack and Taubinsky (2021) for a complete discussion.

2022; Collis and Eggers, 2022) and on the adoption of self-control tools (Hoong, 2021). This paper contributes by providing revealed-preference field evidence based on real monetary choices. The setting allows me to quantify willingness to pay, distinguish high demand from distorted demand, and evaluate counterfactual policies.

More broadly, the paper contributes to the literature on addiction across different contexts. By distinguishing habit formation from self-control problems, I contribute to the debate over whether addiction is rational (e.g., Spinnewyn, 1981; Becker and Murphy, 1988; Orphanides and Zervos, 1995) or driven by behavioral distortions (e.g., Gruber and Köszegi, 2001). The paper also complements studies of addiction in other markets, including cigarettes (Chaloupka, 1991; Becker, Grossman, and Murphy, 1994; Giné, Karlan, and Zinman, 2010), alcohol (Cook and Moore, 2002; Baltagi and Geishecker, 2006), drugs (Gul and Pesendorfer, 2007; Maclean, Mallatt, Ruhm, and Simon, 2020), and sugar-sweetened beverages (Zhen, Wohlgenant, Karns, and Kaufman, 2011).

Finally, this paper contributes to the economics of entertainment goods and digital platforms. Recent studies document key features of the short-video industry, including its addictiveness (Gao, Gao, Meng, and Yu, 2024) and content-supply dynamics (Xia, 2025). The context of short videos also relates to studies of other entertainment goods, such as movies (Michalopoulos and Rauh, 2024), cable television (Crawford and Yurukoglu, 2012; Crawford, Lee, Whinston, and Yurukoglu, 2018), and video games (Lee, 2012). In addition, my work contributes to the literature on social media and digital platforms (e.g., Allcott, Braghieri, Eichmeyer, and Gentzkow, 2020; Liu, Sockin, and Xiong, 2020; Aridor, Jiménez-Durán, Levy, and Song, 2024; Beknazar-Yuzbashev, Jiménez-Durán, and Stalinski, 2024). The contribution is to show how a specific design feature—short content length—can amplify self-control problems and generate quantitatively meaningful welfare losses.

The remainder of the paper is organized as follows. Section 2 presents the stylized model. Section 3 describes the institutional background and reduced-form evidence. Section 4 introduces the structural demand model, whose estimation is discussed in Section 5. Section 6 presents the welfare analysis. Section 7 extends the analysis to TikTok. Finally, Section 8 concludes.

2 Stylized Model of Temptation

I present a stylized model of temptation to demonstrate how could short-form contents amplify self-control problems. I later extend this model into an empirically relevant framework for short drama series in section 4.

2.1 Setup

Consider a platform offering a large set of videos $\{1, \ldots, T\}$, each lasting one minute. Users make decisions at the video level, watching sequentially and deriving a constant intrinsic utility $x \in \mathbb{R}$ from each. I assume for now that users observe x and defer uncertainty and learning to the structural model. The outside option is normalized to 0. Additionally, users experience a *temptation* to watch such videos, valued at $\kappa > 0$ per minute for a duration of $\chi \in \mathbb{N}$ minutes, which could be microfounded by present bias (e.g., Laibson, 1997), love for novelty (e.g., Modirshanechi, Lin, Xu, Herzog, and Gerstner, 2025), inertia, or other behavioral mechanisms. Let the superscript t be the *current video* when the user is making a decision, and let subscript $\tau \geq t$ denote the *evaluated video*, i.e., the video whose utility is being assessed from the perspective of current video t. The perceived utility from video τ at current time t is thus:

$$\widetilde{u}_\tau^t(x, \chi) = \underbrace{x}_{\text{intrinsic utility}} + \underbrace{\mathbb{1}\{\tau < t + \chi\}\kappa}_{\text{temptation } \kappa \text{ with duration } \chi}. \tag{1}$$

For example, starting at current video t, a user with $\chi = 0$ perceives true utility x from each subsequent videos. In contrast, a user with $\chi = 2$ perceives a biased utility of $x + \kappa$ from videos t and $t+1$, and the true utility x from all remaining videos. For this section, I focus on users with $x \in (-\kappa, 0)$, who inherently dislike the short videos relative to outside options but may still choose to watch them due to temptation.

This temptation framework leads to a systematic overweighting of immediate entertainment relative to long-run utility. Depending on the temporal perspective t, users may perceive the same video τ differently, often overvaluing those in the near future (i.e., when $\tau < t + \chi$). For instance, Table 1a illustrates the preferences of users with $\chi = 2$. Before watching the first video ($t = 1$), they perceive the true utility $\widetilde{u}_3^1 = x$ for the third one ($\tau = 3$), because it lies outside their temptation horizon. However, at $t = 2$, the temptation effect emerges, biasing their perception to $\widetilde{u}_3^2 = x + \kappa$. This model therefore generates similar predictions to the quasi-hyperbolic model of present bias from Laibson (1997) and Gruber and Köszegi (2001), but it fits naturally with the discrete consumption of short videos and yields simpler estimation with the separably additive temptation (Banerjee and Mullainathan, 2010; Allcott, Gentzkow, and Song, 2022).[7]

The contribution of this paper is to examine the *duration* of temptation. When interpreting temptation as present bias, this duration captures the length of the "present" as perceived by a present-biased agent.[8] In section 2.3, I show the short format shrinks agents' decision windows below the temptation duration, thereby amplifying the resulting self-control problem. In my structural analysis, I estimate the distribution of temptation durations using variation induced by a non-linear pricing scheme, providing novel empirical insights.

Finally, my analysis focuses on naïve users who are unaware of the time-inconsistency in their preferences. At any given current video t, they mistakenly believe their future preferences will mirror their current ones, failing to recognize their tastes will shift as they continue watching. In section 3.5, I show empirical evidence on self-control problems, which aligns well with the model prediction based on this naïveté assumption. I further discuss this assumption in section 4.4 and characterize the behavior of sophisticated users in Appendix A.2.

[7] In Appendix A.1, I show how to represent the preference (1) from the quasi-hyperbolic discounting model, which micro founds this irrational temptation.

[8] See section 5.1 of DellaVigna (2018) for a detailed discussion regarding the duration of present bias.

Table 1: Example: Self-control problems with $\chi = 2$ and $x \in (-\kappa, -\kappa/2)$

(a) Baseline

Evaluated video τ	Current video t: 1	2	3	4
1	$\boxed{x+\kappa}$			
2	$\boxed{x+\kappa}$	$\boxed{x+\kappa}$		
3	x	$\boxed{x+\kappa}$	$\boxed{x+\kappa}$	
4	x	x	$\boxed{x+\kappa}$	$\boxed{x+\kappa}$
5	x	x	x	$\boxed{x+\kappa}$
6	x	x	x	x

(b) Comparative static: Video length $n = 4$

Evaluated video τ	Current video t: 1	2	3	4
1	$4x+2\kappa$			
2	$4x$	$4x+2\kappa$		
3	$4x$	$4x$	$4x+2\kappa$	
4	$4x$	$4x$	$4x$	$4x+2\kappa$
5	$4x$	$4x$	$4x$	$4x$
6	$4x$	$4x$	$4x$	$4x$

Notes: The tables provide examples of a naïve user with intrinsic value $x \in (-\kappa, -\kappa/2)$ and temptation duration $\chi = 2$. The perceived utility $\tilde{u}^t_\tau$ from watching video τ at perspective t is displayed in the table, with each column representing a current video and each row an evaluated video. Positive utilities are marked with a box, and the highlighted box indicates the videos the user actually watches. Panel (a) illustrates the baseline scenario where the user plans to watch two videos at every decision point but ends up watching all videos. Panel (b) depicts the case with four-minute videos, where the user perceives a flow utility of $4x + 2\kappa < 0$ for each video and thus chooses not to watch.

2.2 Self-control problems

The stylized model generates self-control problems that are summarized in Proposition 1. At current time t, agents with $x \in (-\kappa, 0)$ predict that they will only watch the next χ videos ($\tilde{u}^t_\tau = x + \kappa > 0$) and then stop ($\tilde{u}^t_\tau = x < 0$). However, when they reach video $t + \chi$, they are once again tempted to continue watching for another χ videos, revealing a misprediction and self-control problem: although they intend to stop ex ante, they fail to follow through ex post. Table 1a illustrates an example where a user always plans to watch two videos ($\chi = 2$) at each point in time, but ultimately ends up consuming all the videos provided.

Proposition 1 (Self-control problems) *Users with intrinsic value $x \in (-\kappa, 0)$ and temptation duration $\chi \geq 1$ mispredict future consumption and exhibit self-control problems. At any time t, they predict that they will consume only χ additional videos, but ultimately consume all available videos.*

This self-control problem induces surplus loss on the user side. In the benchmark case without temptation ($\kappa = 0$), users watch the videos if and only if they derive positive intrinsic utility $x \geq 0$. Let (x, χ) represent a user, and let $S \subset \mathbb{R} \times \mathbb{N}$ be the set of users who have self-control problems. The *average surplus loss due to temptation* from short videos can be defined as

$$\Delta(S) = -\sum_{\chi \in \mathbb{N}} \left[T \int_{\{x:(x,\chi)\in S\}} x f_{x|\chi}(x)\, \mathrm{d}x \right] P(\chi), \tag{2}$$

where $P(\chi)$ is the probability mass of χ and $f_{x|\chi}(x)$ is the conditional probability density of x. Proposition 1 yields $S^* = (-\kappa, 0) \times \mathbb{N}_+$ for this baseline case, resulting in a positive user surplus loss $\Delta^* := \Delta(S^*) > 0$.

2.3 Short length exacerbates self-control problems

To understand how short video length exacerbates self-control problems, I consider a hypothetical scenario where the platform extends each video to n minutes.[9] Assume users still make decisions at the video level. The perceived flow utility is now defined as:

$$\widetilde{u}_t^t(x, \chi; n) = nx + \min\{n, \chi\}\kappa, \tag{3}$$

where intrinsic value scales with video length n, and temptation is adjusted by the lesser of video length n or temptation duration χ. If the video length is shorter than the temptation duration ($n \leq \chi$), the perceived utility increases proportionally. However, when the video length exceeds the temptation duration ($n > \chi$), users anticipate losing interest before the video ends, which effectively dampens the influence of temptation.

Given length n, users will watch the videos iff they perceive a positive flow utility, that is, $\widetilde{u}_t^t(x, \kappa, \chi; n) \geq 0$. Self-control problems can thus be characterized by the following condition:

$$\max\left\{-\kappa, -\frac{\chi\kappa}{n}\right\} < x < 0. \tag{4}$$

The self-control problem affects all users with $x \in (-\kappa, 0)$ when the video length n is shorter than the temptation duration χ. However, when $n > \chi$, users with low valuations for the videos ($x < -\chi\kappa/n$) will avoid watching. As n increases, the cutoff $-\chi\kappa/n$ approaches zero, indicating the self-control problem gradually diminishes with video length. Table 1b provides an example with $\chi = 2$ and $n = 4$, where users with $x < -\kappa/2$ escape from the self-control problem.

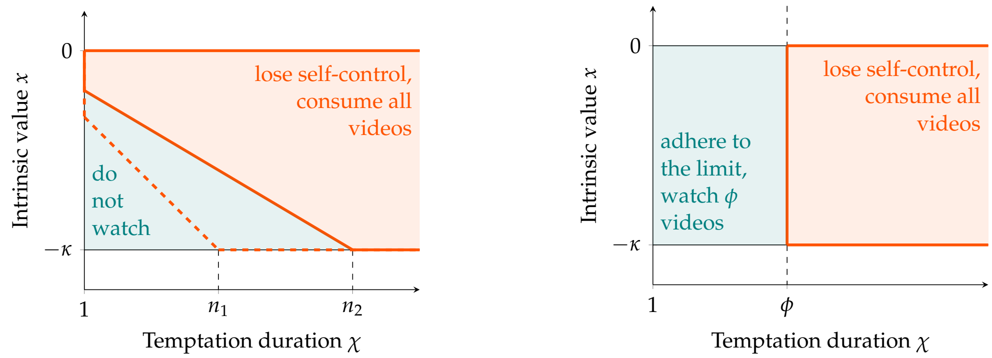
As a result, increasing video length mitigates welfare loss due to temptation. Let S_n^N be the set of users with self-control problems for a given length n, which is characterized by condition (4) and graphically represented by the orange area of Figure 1a. Let $\Delta_n^N := \Delta(S_n^N)$ be the associated average surplus loss due to temptation. For any n_1 and n_2 such that $1 \leq n_1 \leq n_2$, we have:

$$S^* = S_1^N \supset S_{n_1}^N \supset S_{n_2}^N \quad \Rightarrow \quad \underbrace{\Delta(S^*)}_{\Delta^*} \geq \underbrace{\Delta(S_{n_1}^N)}_{\Delta_{n_1}^N} \geq \underbrace{\Delta(S_{n_2}^N)}_{\Delta_{n_2}^N}.$$

When n increases, more users are exempt from the self-control problem, reducing the surplus loss due to temptation defined in equation (2). Proposition 2 summarizes this comparative static.

Proposition 2 (Shorter format exacerbates self-control problems) *Shorter videos engage more users in the self-control problem, leading to greater surplus loss due to temptation. Formally, for any two video lengths n_1 and n_2 such that $1 \leq n_1 \leq n_2$, it holds that $S_{n_1}^N \supset S_{n_2}^N$ and $\Delta^* \geq \Delta_{n_1}^N \geq \Delta_{n_2}^N$.*

[9] This exercise analyzes the impact of video length on self-control problems by merging n one-minute clips into a single, longer video. A real-world analogue is the U.S. law on "loosies," which prohibits the sale of individual cigarettes and mandates a minimum pack size of 20. Such a policy mitigates the effect of momentary temptation to smoke "just one more" cigarette and compels buyers to more fully evaluate the overall value of consumption.



(a) Longer video length n (b) Default time limit of ϕ minutes

Figure 1: Illustration: User behavior under counterfactual scenarios

Notes: This figure illustrates user behavior under various counterfactual scenarios, where users are characterized by intrinsic value (x) and temptation duration (χ). Panel (a) presents the comparative static for video length (n) adjustments. Users with self-control problems, depicted in the orange region, watch all videos despite having a negative intrinsic value, whereas users in the teal triangular area choose not to watch any videos. Panel (b) shows user activity under a default screen time limit of ϕ minutes. The orange region represents users who continue watching all videos, whereas the teal area highlights users who adhere to the limit and watch less videos.

2.4 Counterfactual policy

To address the inefficiency caused by temptation, I consider a counterfactual policy that imposes a default screen time limit of ϕ minutes, allowing users to opt out ex ante but not ex post.[10] To break ties, I assume users must pay a small cost ε to opt out. The user response is illustrated in Figure 1b. For users with $\delta \in (-\kappa, 0)$, those with temptation duration $\chi \leq \phi$ will adhere to the limit because they only plan to watch for χ minutes, whereas those with $\chi > \phi$ expect to watch more and thus choose to opt out. The policy improves consumer surplus by mitigating self-control problems among agents with relatively short temptation durations.

Given user behavior under this policy, I define the average surplus loss due to temptation as:

$$\Delta_{\phi}^{TL} := -\sum_{\chi \leq \phi} \left[\phi \int_{-\kappa}^{0} x f_{x|\chi}(x), \mathrm{d}x \right] P(\chi) - \sum_{\chi > \phi} \left[T \int_{-\kappa}^{0} x f_{x|\chi}(x), \mathrm{d}x \right] P(\chi), \tag{5}$$

where users with $\chi \leq \phi$ consume only ϕ minutes of short videos. The baseline corresponds to both $\phi = 0$ (no compliers) or $\phi = T$ (non-binding limit), in which case $\Delta_0^{TL} = \Delta_T^{TL} = \Delta^*$. The improvement in surplus is given by:

$$\Delta^* - \Delta_{\phi}^{TL} = \sum_{\chi \leq \phi} \left[(T - \phi) \int_{-\kappa}^{0} x f_{x|\chi}(x), \mathrm{d}x \right] P(\chi) > 0, \quad \forall \phi \in (0, T) \tag{6}$$

highlighting a trade-off in choosing ϕ: increasing ϕ expands policy coverage (i.e., the set $\chi \in (0, \phi]$), but reduces the surplus gain per complier, since each watches more content. Proposition 3

[10] A similar policy was implemented by TikTok in 2023, setting a default 60-minute daily limit for users under age 18, with the option to opt out.

formalizes these insights.

Proposition 3 (Default time limit reduces self-control problem) *For users with self-control problems as defined in Proposition 1, introducing a default time limit of $\phi \in (0, T)$ minutes reduces video consumption to ϕ minutes for those with $\chi \leq \phi$, thereby increasing consumer surplus. The optimal value of ϕ balances a trade-off between broader policy coverage and the surplus gain per complier.*

3 Data and Institutional Background

I take the short-drama-series industry as the empirical context for studying the self-control problem and its interaction with content length. I introduce the short drama industry in section 3.1 and provide details about the platform of my study in section 3.2. I then document an aggregate fact across all dramas on the platform in section 3.3: shorter episodes are associated with higher completion rates. The individual-level data for my analysis are described in section 3.4, which I use to establish empirical evidence on misprediction and self-control problems in section 3.5.

3.1 Short drama series

Short drama series are a mixture of drama series and short-form videos. They typically have scripts adapted from web novels and are optimized for vertical viewing on smart phones.[11] The most prominent feature of those mini-dramas is their *shortness*, with each episode of a 40- to 100-episode series lasting about one minute. With a genre similar to soap operas, these short dramas, described as "high on drama, low on glam, and full of plot twists," are designed to get viewers hooked fast, which makes this industry an ideal context for studying temptation and the resulting self-control problem.

Demand for short drama series is high and growing rapidly. In China—the origin of this industry in early 2022—the short drama market generated \$7.0 billion in revenue in 2024, surpassing box office sales of \$5.9 billion.[12] The trend is quickly expanding globally. For example, global short drama platforms such as DramaBox, ReelShort, and ShortMax respectively ranked 6th, 18th, and 33rd in the U.S. Apple App Store Entertainment Top Charts for free apps on April 26, 2024.[13] Users are also willing to pay for these dramas. For instance, the superstar drama series on the platform I study reached \$3.5 million revenue in one month after its first release.

Yet, the production of these short dramas is generally low in quality. The scripts are usually bought and adapted from popular web novels, which typically emphasize immediate engagement over depth and complexity. The plots often rely on sensational or emotionally charged themes

[11] Because of its vertical display mode, this type of series is sometimes also called "vertical drama."

[12] Source: *2024 China Short Drama Industry Research Report*, by Shenzhen Media Group, the Audiovisual Art Research Center of the Communication University of China, and the China Television Drama Production Industry Association.

[13] For comparison, the Apple App Store Entertainment Top Charts on the same day had other related apps such as TikTok (2nd), Netflix (5th), YouTube TV (16th), and AMC Theatres (46th).

such as romance, conflict, or dramatic twists to maintain viewer interest. Character development and nuanced storytelling are minimal, because the focus is on creating visually appealing and attention-grabbing content that can be easily consumed on mobile devices. Short dramas feature relatively unknown actors, avoiding A-list or B-list celebrities, and are directed and edited by mid-level or inexperienced professionals. As a result, production costs are low, mostly below $80,000 per drama, with the entire production cycle completed in seven to 10 days.

The short-drama-series industry thus shares key characteristics with platforms such as TikTok and other short video formats, where low-cost content is designed to be rapidly consumed and highly engaging, often at the expense of quality or depth. This industry offers a unique opportunity to study consumer behavior in the context of short-form video consumption.

3.2 The platform

The data for my study are provided by one of the leading short drama platforms in the industry. I outline the basic structure of this platform in Figure 2, where activities occurring on the platform are highlighted in the shaded area. Dramas are the platform's primary assets. On the supply side, scripts are purchased and adapted from web novels and then produced into short dramas. The platform can acquire existing dramas from external producers through a negotiated two-part tariff or act as an integrated producer to create dramas itself.[14] To attract users, the platform constantly advertises its dramas on major social media platforms such as TikTok and Facebook. Users need to unlock each drama episode by purchasing tokens, subscribing, or watching external ads, which form the platform's three main revenue streams. Below, I provide more details on the components relevant to my study. A more comprehensive description of the platform is available in Appendix B.

User demographics. This platform operates in global markets. The daily active users (DAU) averaged 120,000 in 2024, with the US contributing 23% of users and 63% of revenue. I therefore focus on the US market. Additionally, 86% of the user base is female, and the majority are from younger generations: 36% of users are under 30, 63% are under 40, and 85% are under 50.[15] User overlap with Facebook and TikTok, the two primary social media platforms on which this platform advertises, is substantial, suggesting this sample is representative of a wider audience susceptible to social media addiction.

The (superstar) drama. By March 2024, the platform offered over 400 dramas in all major languages. Adapted from web novels, these short dramas feature dramatic genres, with the most

[14] The two-part tariff includes an upfront payment and a revenue-sharing rule from the platform.

[15] The platform calculated the age for the period between March 1, 2024, and May 14, 2024. The app is labeled "18+," so only users aged over 18 are included. The age distribution may skew even younger because teenagers may access the platform using their parents' phones.

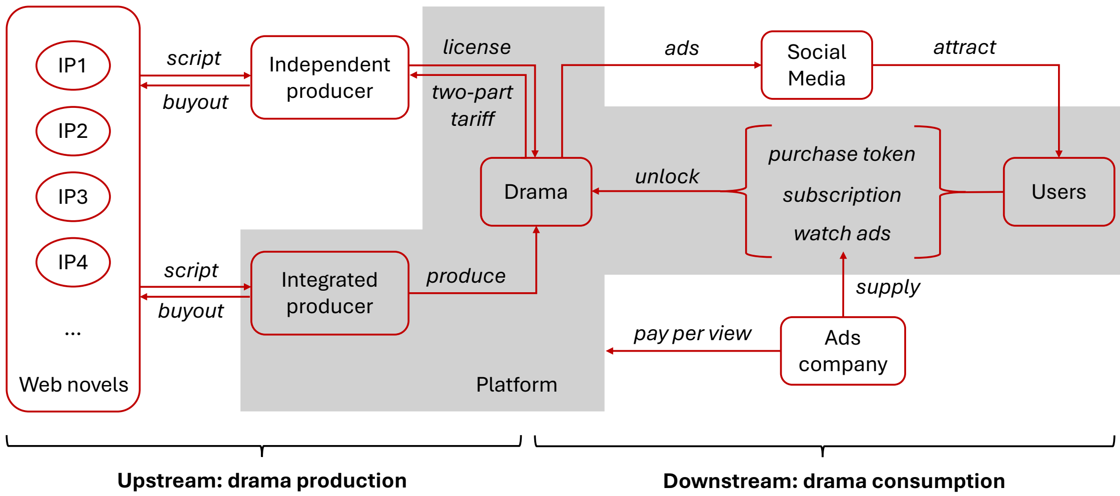


Figure 2: Institutional background of the platform

Notes: The figure summarizes the core business model of the platform. Activities occurring on the platform are highlighted in the shaded area, with short dramas serving as the platform's primary assets. On the supply side, scripts are acquired and adapted from web novels, then produced into short dramas—either by the platform itself or by independent producers. In the latter case, the platform secures licensing rights through an upfront payment and a revenue-sharing agreement. On the demand side, the platform actively promotes its dramas on major social media platforms such as TikTok and Facebook to acquire users. Users unlock each episode by purchasing tokens, subscribing, or watching external ads—these constitute the platform's three main revenue streams. Ads are provided by advertising firms, which pay the platform based on viewership.

popular being "love after marriage" and "toxic love."[16] On average, 1.7 million episodes are consumed daily, with a high level of concentration where a single "superstar" drama accounts for 58.8% of viewership. Spillover across dramas is minimal, with only 3.6% (or 1.5%) of users who have unlocked the superstar drama also unlocking (or topping up for) another drama. Therefore, my main analysis focuses on user behavior related to this superstar drama, which was introduced in mid-January 2024. Consisting of 80 episodes, each lasting one minute, the first nine episodes are free to watch, whereas each subsequent episode costs 65 platform tokens. Produced in-house by the platform, this drama generated $3.5 million in revenue within a month, despite having a production cost of just $70,000. Its success can also be attributed to significant advertising efforts, with the platform investing over $600,000 in promotions within a month.

Drama demand. The payoff-relevant user activity for the platform is unlocking episodes, which can be done in three ways, as summarized in Figure 2: paying tokens, subscribing, or watching ads. My analysis focuses on tokens, the primary revenue source, generating $31,028 in daily income and accounting for 78.0% of total revenue between May 2023 and May 2024.[17] Users must

[16]The "love after marriage" genre explores romantic relationships that develop and evolve post-marriage. "Toxic love" refers to a dysfunctional relationship characterized by unhealthy behaviors, such as manipulation, control, emotional abuse, and dependency.

[17]The second-largest revenue source is subscriptions, a model commonly used by drama platforms such as HBO. The platform offers three subscription options varying by duration: a week ($29.99), a month ($59.99), and a year ($199.99), contributing an average daily income of $6,631 and 16.7% of total revenue. The final revenue stream comes

unlock each episode sequentially by spending a specified amount of platform tokens, providing ideal variation to identify willingness to pay on an episode-by-episode basis.

Additionally, users must purchase top-up packages with real money when they run out of tokens.[18] The platform offers four packages: $4.99 for tokens equivalent to 7.50 episodes, $9.99 for 16.62 episodes, $19.99 for 36.74 episodes, and $29.99 for 81.49 episodes. Random promotions introduce some variation in the number of tokens for each package across users and over time. As shown in Table 4, the platform employs a non-linear pricing structure, where the average price per episode decreases with larger package sizes. This structure creates a trade-off between lower costs and greater commitment, providing a source of identification for users' ex-ante preferences. The top-up page appears whenever users deplete tokens, and payment can be made seamlessly through the App Store or Google Play within seconds, making the process nearly frictionless.

User behavior. After unlocking an episode, users may rewatch it an unlimited number of times. However, I focus on the initial viewing of each episode and do not account for repeated viewing behavior. Users in the sample are required to unlock episodes sequentially, and they typically watch an episode in its entirety at least once before unlocking the next. This observation supports the assumption that users make viewership decisions at the video level.

3.3 Motivating fact: Shorter episodes predict higher completion rates

Before turning to user-level choices in the microdata, I use aggregate data from all drama series on the platform to document a motivating fact about content length. The outcome of interest is the drama-level completion rate, defined as the fraction of first-episode viewers who reach the final episode of a series. I construct this measure by matching the platform's drama metadata to episode-level retention data. The metadata include each drama's average episode length, number of episodes, language, release date, genre tags, and video bitrate. The retention export reports, for each drama and episode, the retention rate and viewer count.[19] I restrict the sample to dramas with positive first-episode viewership, language information, and non-missing completion rates. The resulting sample contains 353 drama series. Average episode length in this sample has mean 1.41 minutes and standard deviation 0.28 minutes.

I estimate the following sequence of drama-level reduced-form regressions:

$$\text{CompletionRate}_d = \beta \text{Length}_d + \mathbf{X}_d' \Gamma + \text{Fixed Eeffects}_d + \varepsilon_d, \tag{7}$$

where Length_d is the average episode length, in minutes, for drama d. The specifications move se-

from ads, where users watch 30-second ads to unlock episodes, earning the platform \$4,329 in daily income and 5.4% of total revenue. The time series of these income sources is reported in Appendix B.

[18] As part of the platform's advertising strategy, users can also earn a small amount of gift tokens through daily log-ins, sharing content on social media, and following the platform's TikTok account, but these amounts are negligible compared with top-up packages.

[19] The retention data cover viewing between March 1 and March 31, 2024, and the matched dramas in the analysis were released between November 10, 2022, and March 15, 2024.

Table 2: Episode length and drama completion rate (%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Episode length (minutes)	−2.18	−2.75**	−3.22**	−2.79**	−2.58**	−2.46***	−2.42***
	(1.52)	(1.14)	(1.28)	(1.25)	(1.02)	(0.92)	(0.92)
Fixed effects							
Language		Yes	Yes	Yes	Yes	Yes	Yes
Genre			Yes	Yes	Yes	Yes	Yes
Release-month				Yes	Yes	Yes	Yes
Controls							
Episode count					Yes	Yes	Yes
First-episode viewers						Yes	Yes
Bitrate							Yes
Observations	353	353	353	353	353	353	353
Mean dep. var.	11.44	11.44	11.44	11.44	11.44	11.44	11.44
R^2	0.00	0.73	0.74	0.77	0.78	0.80	0.80

Notes: The dependent variable is the drama-level completion rate, defined as the fraction of first-episode viewers who remain until the terminal episode. The coefficient on episode length is measured in percentage points per additional minute. Standard errors in parentheses are HC1 robust. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

quentially from broad composition controls to richer series-level controls. Starting from a length-only specification, I add language fixed effects, genre fixed effects, and release-month fixed effects, which absorb differences in completion rates driven by market/localization, content category, and release-cohort conditions, respectively. Other controls include episode count, first-episode viewers, and video bitrate, which control for the scale of the series, initial audience size, and technical quality, respectively. This exercise is descriptive: episode length is chosen jointly with other content, production, and marketing decisions, so the estimates should be interpreted as conditional correlations rather than causal effects.

Table 2 documents a robust negative relationship between episode length and completion. Column (1) reports the raw correlation: an additional minute of episode length is associated with a 2.18-percentage-point lower completion rate. Columns (2)–(4) add language, genre, and release-month fixed effects. The point estimate becomes larger in magnitude after accounting for language and genre composition, and remains negative after absorbing release-month differences across dramas. Columns (5)–(7) add episode count, first-episode viewers, and bitrate. The coefficient remains stable across these richer specifications. In the full specification, the coefficient is −2.42 percentage points per minute, with a standard error of 0.92. Equivalently, shortening the average episode by 30 seconds is associated with a 1.21-percentage-point higher completion rate. Relative to the mean completion rate of 11.44%, this amounts to 10.6% of the mean and is therefore economically meaningful.

This aggregate fact motivates the central mechanism of the paper. Across dramas on the same platform, shorter episodes are associated with a higher probability of completing the series. This pattern is consistent with the idea that short content lowers the stopping margin: each episode requires only a small incremental commitment, and the decision to stop is repeatedly deferred to the next short interval. The aggregate evidence does not, by itself, identify either a causal effect of episode length or a self-control mechanism. It does, however, show that the content-length margin emphasized in the model is visible in platform-wide data. The user-level analysis below then studies the behavioral mechanism behind this relationship, using top-up and unlocking choices to distinguish rational continuation from self-control problems.

3.4 Individual-level data and summary statistics

For a representative, randomly selected 10% sample of global users, I have access to individual-level log data that encompass users' full activities, including top-ups, unlocking, subscriptions, and viewership, from November 1, 2023, to March 31, 2024.[20] I focus on two key dimensions: top-up package purchase and unlocking. Each time users purchase platform tokens, I observe the transaction time, the price paid, and the number of tokens acquired. Additionally, I record the time, method, and number of tokens spent to unlock each drama episode.[21] This information fully characterizes users' drama-watching activities via platform tokens.

Individual-choice data. I outline the data cleaning procedure below and provide full details in Appendix C.1, including the rationale and resulting sample size for each step, and a discussion of potential selection introduced throughout the process. Based on institutional knowledge, I focus on top-up and unlocking activities related to the superstar drama by users located in the U.S. I exclude users who unlocked another drama before or during their engagement with the superstar drama, as well as those who accessed episodes via subscriptions or ad viewing. To focus on regular usage patterns, I drop users who did not watch the episodes in their natural sequential order. To ensure users had sufficient time to complete the drama, I retain only those who began watching before January 31, 2024, and exclude users who had not finished the series but remained active during the final week of the sample period (March 24–31, 2024). To ensure that tokens were primarily acquired through top-up purchases, I exclude users whose initial or gift token endowment exceeded the equivalent of five episodes.[22]

For this sample, I define a "round" as a sequence in which any two consecutive episodes are unlocked within two hours, capturing the behavior of users who pause and later resume viewing.[23] I further refine the sample by excluding outliers in the top percentile of the round distri-

[20] Each user is randomly assigned a unique user ID upon first opening the platform app. Users in my sample are those whose user ID ends with the number "8," effectively creating a 10% random sample of all users.

[21] For free episodes, I consider the first-time viewership as the equivalent unlocking activity.

[22] Users could obtain initial or gift tokens through platform promotions such as daily log-ins or sharing the drama on social media. I exclude users whose non–top-up token value exceeds five episodes, which accounts for 8.7% of the sample. This small share suggests that most users did not rely heavily on such tokens, and that the exclusion assumption is unlikely to affect the quantitative results. A full discussion, including the distribution of initial and gift tokens, is provided in Appendix C.1.

[23] I model round formation as a stochastic and exogenous process, introduced in the structural model. This simplifying assumption abstracts from pausing behavior, which is not central to the analysis. Section 4.4 discusses round dynamics in greater detail, including the underlying rationale, robustness checks, and potential biases. All results

Table 3: Summary statistics on the user level

Variable	mean	std	min	p25	p50	p75	p90	p95	p99	max	count
Episode	19.03	23.79	1	9	9	12	80	80	80	80	11,512
Top-up ($)	5.80	13.35	0	0	0	0	29.99	39.98	49.97	69.96	11,512
Round	1.37	0.73	1	1	1	2	2	3	4	5	11,512
Gift	0.29	0.96	0	0	0	0	1	2	5	12	11,512
Endowment	0.43	0.91	0	0	0	0	2	3	4	4	11,512

Table 4: Top-up packages: Tokens and market share

Price ($)	tokens	[p5,p95]	price/episode	market share
4.99	7.50	[7,9]	0.67	0.34
9.99	16.62	[16,20]	0.60	0.36
19.99	36.74	[36,46]	0.54	0.20
29.99	81.49	[62,92]	0.37	0.11

bution, users who purchased top-up packages not listed in Table 4, and those whose packages contained an unusually rare token quantity (frequency < 5%, conditional on prices).

This data-cleaning process results in a sample of 219,064 unlocking events and 5,360 top-up activities by 11,512 users. For ease of interpretation, I normalize all token quantities by the per-episode cost in the analysis that follows.

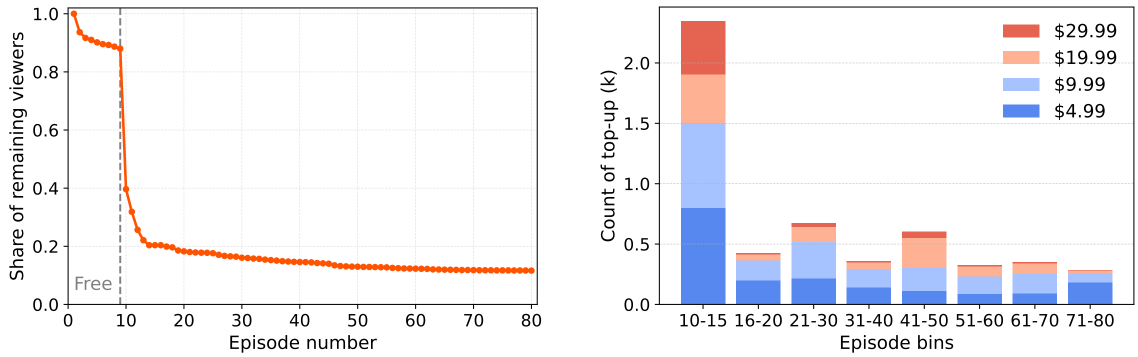
Summary statistics. Summary statistics for these users are presented in Table 3. On average, a user unlocks 19.03 episodes and spends $5.80 on this drama. Considerable heterogeneity exists, with over three-quarters of users not spending anything and only unlocking free episodes. More than half users consume the videos in a single round. The average value of gift tokens and initial endowments per user corresponds to 0.29 and 0.43 episodes, respectively—both small relative to top-up purchases.[24] For simplicity, I model these gifts as extra tokens included with all the top-up packages in my quantitative analysis.

Table 4 presents summary statistics related to top-up choices. The platform employs non-linear pricing, where the average price per episode decreases with higher-price packages. This structure highlights the trade-off between a lower average cost and a larger ex-ante commitment in the top-up decision. All four packages have a positive market share, allowing for the identification of users' ex-ante preferences and temptation durations based on observed top-up choices through a revealed-preference argument. Additionally, I observe variation in the number of tokens across top-up packages due to promotional offers randomly presented to users upon accessing the top-up page, which facilitates the identification of price sensitivities.

Figure 3a presents the share of unlocking activity across episodes. By construction, all 11,512

remain robust to defining rounds using different (e.g., 12-hour) thresholds.

[24] Users can receive free tokens as rewards for engaging in platform activities such as daily logins or sharing content on social media. The quantity of these gift tokens is small, rendering such promotions negligible for my analysis.



(a) Share of remaining viewers by episode

(b) Package purchases by episode bins

Figure 3: The dynamic patterns in unlocking and top-up package purchasing decisions

Notes: Panel (a) reports the share of users who remain viewing each episode, which is monotonically decreasing because users have to unlock sequentially. The total number of users is 11,512. The first nine episodes are free to watch, accounting for the sharp drop at the 10th episode. Panel (b) summarizes the dynamic pattern in top-up package sales by bins of five or 10 episodes.

users in my sample watch the first episode. After each episode, some users discontinue watching, and the share of unlocks per episode stabilizes relatively quickly. The most significant drop in viewership occurs at the 10th episode, coinciding with the end of the nine-episode free trial, when users must purchase tokens to continue.[25] Three-quarters of users exit the series without purchasing any tokens. Ultimately, 1,340 users (11.64%) complete the entire drama. Figure 3b illustrates the dynamic pattern of top-up package choices, grouped in bins of five or 10 episodes. The largest package, priced at $29.99, is predominantly purchased between the 10th and 15th episodes by highly engaged users who are willing to commit to watching many episodes. The other three packages exhibit positive sales across all episode bins. The persistent market share of lower-priced packages across episodes suggests a self-control problem: users initially plan to watch only a few episodes ex ante but fail to stop ex post. I further investigate this pattern in section 3.5.

3.5 Evidence of self-control problems

In this section, I examine patterns in user behavior related to self-control problems, which provide essential variation for identifying temptation in section 4. The identification arises from the nonlinear pricing scheme in top-up package purchases. Figure 4a illustrates this idea with an actual user who repeatedly purchases the smallest top-up package and completes the entire drama in a single night.[26] The choice of the $4.99 package suggests the user initially intended to watch only the next seven episodes at each purchase; otherwise, opting for a larger package with a lower per-

[25]The probability of discontinuing peaks at the 10th episode but remains elevated in subsequent episodes, as some users possess a small token endowment (sufficient for one to four episodes) and delay their top-up decision until exhausting their token stock.

[26]The fact that this user finishes the series in one night rules out alternative explanations, such as "budgeting," which might apply if they had spaced out their viewing over several weeks.

episode cost would have been more rational. However, their ex-post behavior—continuing to the end—reveals an inability to commit to stopping, indicating a severe self-control problem.[27] This pattern is widespread in my sample, as reflected in the persistently high market share of small top-up packages throughout the drama in Figure 3b.

To further illustrate time inconsistency, in Figure 4b I compare the distribution of intended versus actual number of episodes watched for users who have purchased tokens. Intended consumption, represented by the orange cumulative density curve, is inferred from users' first token package choice. On average, users initially intend to watch 28.0 episodes when making their first top-up decision. By contrast, the actual number of episodes watched, shown by the red curve, exhibits first-order stochastic dominance over the intended distribution. The average number of episodes actually viewed is 51.0, which is 82.1% higher than users' initial intention. This significant discrepancy—where actual consumption far exceeds intended consumption—provides strong evidence of self-control problems.

Moreover, due to the platform's non-linear pricing scheme, the fact that the example user in Figure 4a could have spent less motivates the use of *overpayment* as an indicator of self-control problems. Figure 4c presents the distribution of overpayment, measured relative to optimal spending, among users who made at least one package purchase.[28] The results indicate 39.1% of these users spend more than the rational benchmark to unlock episodes, suggesting self-control problems are prevalent in my sample. Specifically, 17.6% overpay by approximately \$10, 9.5% by \$15, and 8.2% by \$20. On average, these users overpay by \$5.51, which is 22.7% more than the optimal expenditure, underscoring the substantial impact of self-control problems in this setting.

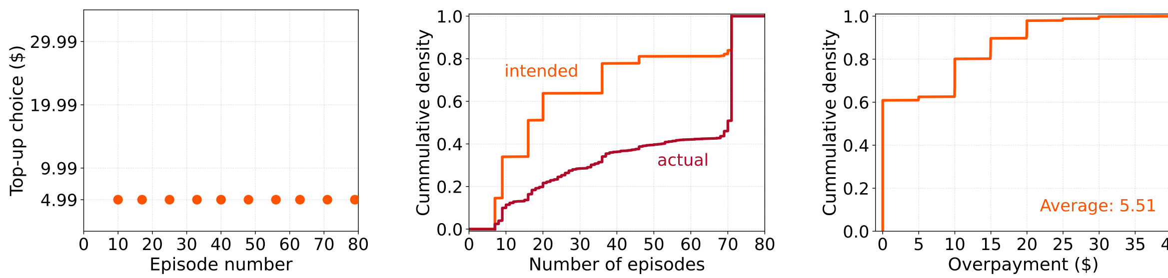
Uncertainty and learning. Learning provides a natural competing explanation for the patterns in Figures 4a–4c. A user who is uncertain about drama quality may initially buy the small package to acquire information, making a \$4.99 purchase a plausible trial choice. As experience accumulates, however, this trial motive should weaken. Favorable signals should make larger packages more attractive, while unfavorable signals should lead users to stop. This implication is sharpened by the martingale property of Bayesian beliefs: a user's current belief is her best forecast of her future belief, so the next signal is not expected, ex ante, to move her valuation systematically in one direction, let alone toward repeated small trials. Learning should therefore both sort users across actions and reduce the incentive to keep experimenting with the smallest package.

Panels (d) and (e) show the opposite pattern. Panel (d) shows that, after purchasing the \$4.99 package, many users continue buying it; moreover, this tendency increases as they watch more episodes and accumulate more information.[29] Panel (e) focuses directly on this repeated-purchase

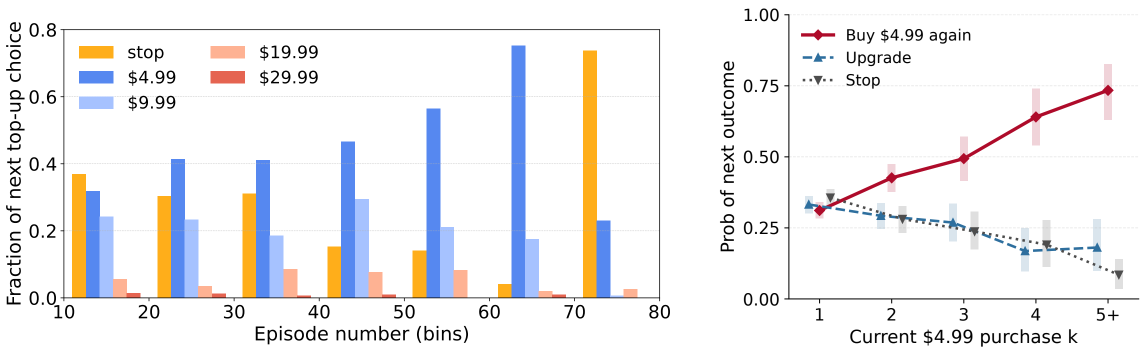
[27] An alternative explanation is that users anticipate a higher likelihood of stopping as they progress further into the series, making them more cautious in token purchases. For example, if a user is waiting for a call, the longer they wait, the more likely they are to receive it in the next few minutes. However, the aggregate viewing pattern does not support this explanation. As shown in Figure 3a and Figure C.5 in Appendix C.4, the probability of continuing to the next episode increases quickly and then remains consistently high throughout the rest of the series.

[28] Optimal spending is imputed using the average token quantity in each package from Table 4.

[29] The spike in stopping behavior between episodes 70 and 80 is mechanically high, because users must stop after completing the series.



(a) Self-control problem example (b) Intended vs. actual usage (c) Distribution of overpayment



(d) Next top-up package choice following $4.99 purchase (e) Repeated $4.99 purchases

Figure 4: Evidence of self-control problems

Notes: Panel (a) illustrates an example of self-control problems, in which the user repeatedly purchases the smallest top-up package. The y-axis lists the available top-up options, and the x-axis indicates the episode before which each top-up occurs. Panel (b) compares the cumulative distributions of intended and actual episodes watched among users who purchased token packages. Intended consumption is inferred from the first package purchased. Panel (c) presents the cumulative distribution of overpayment among users who made at least one package purchase. Overpayment is measured relative to optimal spending, imputed using the average token quantity in each package from Table 4; its natural lower bound is $0. Panel (d) reports users' next actions after a $4.99 purchase, separately by episode group. The panel uses accumulated viewing experience as a proxy for how much uncertainty about drama quality has been resolved. Under a learning-only explanation, users who have watched more episodes should be less likely to keep experimenting with the same small package and be more likely to upgrade or stop as beliefs move upward or downward. Instead, many users continue purchasing the $4.99 package even after substantial viewing experience. The sharp increase in stopping between episodes 70 and 80 is mechanical, because users must stop after completing the series. Panel (e) sharpens the same test by conditioning on the order of repeated $4.99 purchases. It reports the next outcome after the kth $4.99 purchase. Rectangular bars denote 95% confidence intervals from a user-level bootstrap. The probability of another $4.99 purchase rises with k, while the probability of upgrading and stopping falls. This persistence in repeated small purchases provides the empirical moment used to separate temptation from learning.

margin by asking what users do after their *k*th \$4.99 purchase. The probability of buying the \$4.99 package again rises from 31.1% after the first such purchase to 73.4% after at least five such purchases. By the fifth-or-later \$4.99 purchase, only 18.1% of users upgrade and 8.5% stop. This increasing persistence in small-package repurchases is difficult to reconcile with a learning-only account in which experience resolves uncertainty; instead, it points to a persistent local-repurchase force consistent with self-control problems.

In Appendix C.4, I follow DellaVigna and Malmendier (2006) to account conservatively for rational motives and estimate an ex ante lower bound on overpayment due to self-control problems. The exercise counterfactually replaces the first \$4.99 purchase in each episode bin with a larger package, while holding realized viewing fixed and preventing users from reoptimizing. The main takeaway is that late \$4.99 purchases still imply positive lower-bound savings. In the 40–50 and 50–60 episode bins, replacing the first \$4.99 package with the \$9.99 package saves \$0.20 and \$0.30 per user on average, respectively. Because this exercise abstracts from uncertainty, learning, and other rational motives for choosing the smallest package, these positive entries provide conservative evidence of overpayment associated with self-control problems.

The structural model in section 4 therefore allows users to learn about drama quality, but it also targets these conditional transition probabilities in Figures 4e as moments that help disentangle learning from temptation. Learning is disciplined by how users sort into upgrading or stopping, whereas temptation is disciplined by the persistent and increasing tendency to keep purchasing the smallest package.

Underlying mechanisms. In this section, I present evidence of self-control problems while remaining agnostic about the precise mechanisms underlying the observed time inconsistency. In the model, short-term behavioral temptation may be interpreted as present bias that overweights immediate entertainment or an inertia that hinders stopping. Although the data lack the variation needed to distinguish among these mechanisms, they are isomorphic in terms of behavioral predictions and welfare implications. I therefore adopt the general framework of temptation to analyze the problem.

4 Structural model

I adopt a structural approach to systematically examine self-control problems in the short-drama-series industry while allowing users to learn about drama quality. Building on the stylized model from section 2, I develop a single-agent dynamic discrete-choice model in section 4.1. The model solution is characterized in section 4.2. Section 4.3 reports main predictions on self-control problems, followed by a discussion of modeling assumptions and interpretation in section 4.4.

4.1 Setup

A superstar drama consists of T episodes indexed by t. Each one-minute episode defines a decision period in which the user decides whether to purchase a top-up package and unlock the next episode. The drama-watching activity can therefore be modeled as a finite-horizon dynamic discrete-choice problem. This section presents the single-agent setup; I omit the individual subscript i for simplicity.

Welfare-relevant utility. Let the true quality of the drama for this user be fixed at δ per episode, where δ varies across users according to $\mathcal{N}(\mu_\delta, \sigma_\delta^2)$. I define the welfare-relevant utility from consuming episode t as

$$u_t^*(h_t, \delta, \psi_t, \epsilon_t) := \underbrace{\overbrace{\delta}^{\text{quality}} - \overbrace{\psi_t}^{\text{outside}}}_{\text{intrinsic utility}} + \underbrace{\alpha(h_t)}_{\text{habit formation}} + \underbrace{\epsilon_t}_{\text{taste shifter}}, \tag{8}$$

which consists of intrinsic utility, habit formation, and a taste shifter. The intrinsic utility captures the value of the drama relative to the user's outside option ψ_t. The taste shifter ϵ_t captures episode-specific shocks to the user's utility and is unknown to the user before time t.

I allow for *rational addiction* through habit formation (Becker and Murphy, 1988).[30] When encountering episode t, the user has habit stock h_t, defined as the number of previous episodes watched within the current round r_t. This stock depreciates across rounds. Habit stock enters utility through the nonlinear function $\alpha(h_t) = \alpha_1 h_t(h_t - \alpha_2)$.[31] Rounds evolve exogenously: after watching an episode, the user must start a new round with probability ρ, so that $r_{t+1} = r_t + 1$.[32] By design, habit formation is temporary: it accumulates within a round and captures the transient nature of digital addiction. I assume that users have rational expectations about both habit formation and round dynamics, and fully internalize these forces when making drama-watching decisions.

In practice, I assume that the outside value ψ_t is round-specific and depends on the time of day at which the user begins watching. I normalize $\psi_t = 0$ for rounds that start during the day, defined as 11am–12am, and parameterize the outside value for nighttime, defined as 1am–10am, as $\overline{\psi}$.[33]

[30] In drama series, episodes are linked by a continuous storyline, which helps keep users engaged and makes habit formation relevant. Appendix C.2 provides evidence on habit formation by examining the relationship among habit stock, the probability of continuing watching, and the probability of purchasing higher-priced packages.

[31] I adopt this quadratic functional form for the following reasons. First, it satisfies $\alpha(0) = 0$, implying no additional utility in the absence of habit stock. Second, when $\alpha_1 < 0$, it generates an inverted-U shape, capturing diminishing returns due to fatigue. Third, the parameters are straightforward to interpret. This functional form is not critical to my quantitative results.

[32] The assumption of exogenous round dynamics is crucial for identifying habit formation. It is reasonable because if users stop watching because they dislike the series, they should not resume in the next round after losing all accumulated habit stock. Thus, any observed temporary disruptions must stem from exogenous factors—for example, a user might receive an unexpected email about a paper decision, interrupting their drama-watching session. See section 4.4 for a complete discussion of this round dynamic.

[33] This choice of time window is motivated by viewing patterns in the data, which show below-average viewership

The probability that a new round begins at night, $\varphi = 0.27$, is calibrated directly from the data. I also assume that the random taste shifter ϵ_t follows a Gumbel distribution, $Gumbel(-\beta_\epsilon \gamma, \beta_\epsilon)$.[34]

Perceived flow utility. When making choices, however, the user perceives flow utility as:

$$u_t(h_t, \widehat{\delta}_t, \psi_t, \chi, \epsilon_t) = \underbrace{\overbrace{\widehat{\delta}_t}^{\text{belief}} - \overbrace{\psi_t}^{\text{outside}}}_{\text{expected intrinsic utility}} + \underbrace{\mathbb{1}\{\chi \geq 1\}\kappa}_{\text{temptation}} + \underbrace{\alpha(h_t)}_{\text{habit formation}} + \underbrace{\epsilon_t}_{\text{taste shifter}} . \tag{9}$$

The perceived utility differs from welfare-relevant utility in two respects: it includes a short-run, irrational temptation term, and it replaces true drama quality with the user's belief, which evolves through rational learning.

First, following the stylized model in Section 2, I capture self-control problems as short-run temptation with naïve perception. This formulation can be microfounded by behavioral mechanisms such as present-biased preferences or inertia.[35] The per-minute temptation κ is active when its duration χ exceeds the one-minute episode length.[36]

Second, the user does not directly observe the fixed drama quality δ. Instead, before episode t, the user holds a posterior belief with mean $\widehat{\delta}_t$. Users enter the drama with a rational-expectations prior. After watching episode t, the user observes a clean signal $x_t = \delta + \eta_t$, where $\eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$ captures episode-level experience noise and is distinct from the taste shifter ϵ_t. By Kalman filter, the posterior before episode t is $\mathcal{N}(\widehat{\delta}_t, \widehat{\sigma}_t^2)$, where

$$\widehat{\sigma}_t^2 = \left(\frac{t-1}{\sigma_\eta^2} + \frac{1}{\sigma_\delta^2} \right)^{-1} \quad \text{and} \quad \widehat{\delta}_t = \widehat{\delta}_{t-1} + \frac{\widehat{\sigma}_{t-1}^2}{\widehat{\sigma}_{t-1}^2 + \sigma_\eta^2} \left(x_{t-1} - \widehat{\delta}_{t-1} \right). \tag{10}$$

Because the posterior variance is deterministic in the number of watched episodes, the stochastic learning state is summarized by $\widehat{\delta}_t$.

Dynamics. I now formalize the model's dynamic components. To match the data, I assume that all users watch the first episode and enter the drama with prior mean $\widehat{\delta}_1 = \mu_\delta$. I also assume that each user experiences a temptation spell lasting $\chi \in \{0, 1, \ldots, 60\}$ minutes and forms naïve perceptions about temptation when forecasting future choices.[37] The 60-minute upper bound cap-

between 1am and 10am. A detailed description can be found in Appendix C.3.

[34] The parameter β_ϵ is the scale parameter that controls the dispersion of the Gumbel distribution. The location parameter is set to $-\beta_\epsilon \gamma$, where γ is the Euler constant, so that the expectation of ϵ_t is zero (e.g., Rust, 1987).

[35] I remain agnostic about the precise mechanism. These mechanisms are modeling-wise isomorphic, because different interpretations of the temptation term yield the same welfare implications.

[36] For simplicity, I assume that temptation remains constant every minute within its duration, although a more flexible model could allow it to decay over time. Although I cannot estimate the rate at which temptation decays, incorporating this feature would only strengthen the present bias, further amplifying the self-control problem and making the short video length an even greater factor in driving addiction.

[37] As shown in Appendix A.2, sophisticated users who dislike the videos would never engage with the platform. Therefore, I assume all users in my sample are naïve. In Section 4.4, I discuss potential bias from this assumption and

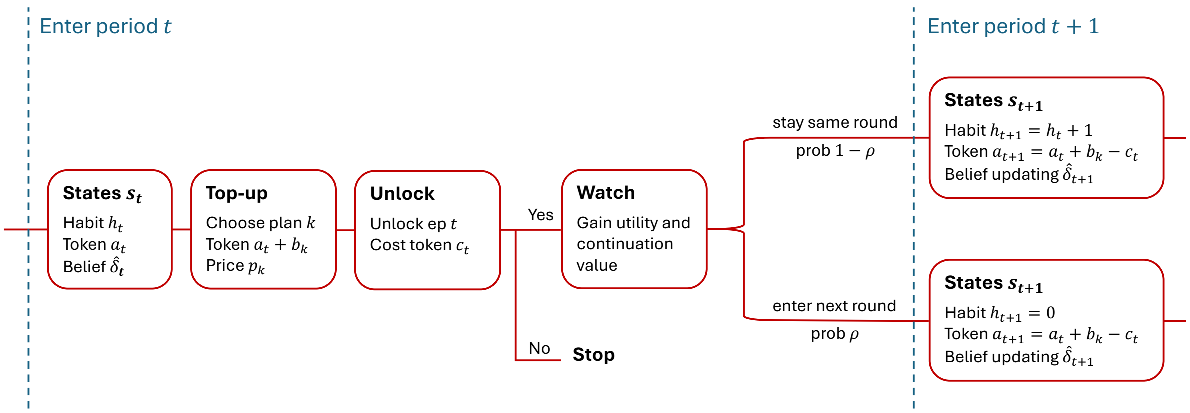


Figure 5: Timing: Decisions on top-up and unlocking within a time period

tures the short-run nature of temptation and improves the tractability of the solution algorithm. At the beginning of each episode $t \geq 2$, the user has unlocked and watched the previous $t-1$ episodes. The platform sets the token cost as $c_t = \mathbb{1}\{t > 9\}$, so users begin paying a normalized token amount after the ninth episode. The state is $\mathbf{s}_t$, which consists of habit stock h_t, token stock a_t, and belief $\widehat{\delta}_t$.[38] Figure 5 summarizes the timing of decisions. The user makes two sequential decisions: whether to purchase a top-up package and whether to unlock the next episode.

First, the user decides whether to top up when the current token stock a_t is below the cost of unlocking the next episode, c_t.[39] I assume that the top-up process is frictionless, because users can complete purchases within seconds.[40] They choose from a menu of $K = 4$ packages, indexed by $k \in \{1, \ldots, K\}$, with price p_k and token quantity b_{kt}, as well as an outside option denoted by $p_0 = b_0 = 0$. Packages are ordered so that p_k increases with k. Token quantities are randomly drawn for each user-period pair from a platform-set distribution $F_{\boldsymbol{b}}(\cdot)$, reflecting random promotions. In practice, the platform uses nonlinear pricing: the average price per token, $\mathbb{E}[p_k/b_{kt}]$, decreases with package price. This pricing structure creates a trade-off between lower per-token prices and greater ex ante commitment to watching additional episodes. Top-up choices therefore reveal users' ex ante preferences over how many episodes they intend to watch, given their short-lived temptation and current beliefs. I also assume an idiosyncratic taste shifter ν_{kt} for each top-up package, distributed as standard Gumbel. I denote the value and expected value functions associated with the top-up decision by $V_t(\mathbf{s}; \psi, \chi, \boldsymbol{b}, \boldsymbol{\nu})$ and $EV_t(\mathbf{s}; \psi, \chi)$, respectively, where state variables appear before the semicolon. Section 4.2 provides the full solution.

Second, after any top-up decision, the user decides whether to unlock episode t, given habit

conclude that including (partially) sophisticated users would further strengthen the case for self-control problems.

[38]The posterior variance $\widehat{\sigma}_t^2$ is pinned down by t and does not need to be carried as a separate stochastic state.

[39]In practice, the top-up page only appears when users have insufficient token balance. Users could, in principle, stop watching, return to the main page, and manually access the top-up page at any time, but fewer than 1% of users choose to do so when they have sufficient tokens.

[40]This assumption is quantitatively conservative because introducing friction would reduce users' willingness to top up, implying that the structural model would estimate an even stronger present bias to account for the time-inconsistent behaviors observed in the data.

stock h_t, token stock $a_t + b_{k^*t}$, and belief $\widehat{\delta}_t$. If the user unlocks the episode, they expect to receive perceived flow utility u_t, defined in equation (9), together with a perceived continuation value, denoted by $\mathcal{C}_t(\mathbf{s};\psi,\chi)$ for notational simplicity. This continuation value comes from the option to unlock episode $t+1$. When evaluating this option, users perceive themselves to be one minute less tempted under the naïveté assumption and form Bayesian expectations over the next belief $\widehat{\delta}'$. I denote the perceived value and expected value functions associated with the unlocking decision by $W_t(\mathbf{s};\psi,\chi,\boldsymbol{\epsilon})$ and $EW_t(\mathbf{s};\psi,\chi)$, respectively.

Finally, I assume that neither habits nor tokens generate continuation value outside this superstar drama. This yields the terminal condition

$$EV_{T+1}(\cdot) \equiv EW_{T+1}(\cdot) \equiv 0. \tag{11}$$

This assumption is motivated by the empirical observation that spillovers across dramas are minimal and that most users do not watch other content after completing this drama.[41] The terminal condition in (11) therefore makes the user's dynamic problem finite horizon.

4.2 Solution: Recursive definition of value functions

Following O'Donoghue and Rabin (1999), I use the concept of *perception-perfect equilibrium strategies* to account for the behavioral forces in my model. Under this framework, users maximize their *perceived* utility, which can lead to irrational self-control problems. I define the value functions iteratively by using backward induction, which, combined with the terminal condition (11), fully characterizes the solution.

For any episode $t \leq T$, the perceived value function associated with the top-up decision for a user with habit stock h, token stock a, posterior mean $\widehat{\delta}$, outside value ψ, token amounts b_k, perceived temptation duration χ, and taste shifter ν_k is given by:

$$\begin{aligned} V_t(\mathbf{s};\psi,\chi,\boldsymbol{b},\boldsymbol{\nu}) = \quad & \mathbb{1}\{a < c_t\} \max_{k\in\{0,1,\ldots,K\}} \left\{ EW_t(h, a+b_k, \widehat{\delta};\psi,\chi) - \omega p_k + \nu_k \right\} \\ & + \mathbb{1}\{a \geq c_t\} EW_t(h,a,\widehat{\delta};\psi,\chi), \end{aligned} \tag{12}$$

where $EW_t(\cdot)$ is the perceived value from unlocking and ω represents price sensitivity. If the token balance is insufficient ($a < c_t$), the user must choose one of the top-up options k; otherwise, they skip topping up and proceed to the unlocking stage. Because the taste shifter ν_{kt} follows a standard Gumbel distribution, the expected value can be written as:

$$\begin{aligned} EV_t(\mathbf{s};\psi,\chi) = \quad & \mathbb{1}\{a < c_t\} \mathbb{E}_{\boldsymbol{b}} \left[\log \left(\sum_{k=0}^{K} \exp \left[EW_t(h, a+b_k, \widehat{\delta};\psi,\chi) - \omega p_k \right] \right) \right] \\ & + \mathbb{1}\{a \geq c_t\} EW_t(h,a,\widehat{\delta};\psi,\chi), \end{aligned} \tag{13}$$

[41] See more details in Appendix C.5.

where the expectation $\mathbb{E}_{\boldsymbol{b}}(\cdot)$ is taken over the token amount $\boldsymbol{b}$.

In the unlocking stage, the user decides whether to unlock episode t. By unlocking, they derive the perceived flow utility $u_t(h, \widehat{\delta}, \psi, \chi, \epsilon)$ and gain the perceived continuation value defined as:

$$\begin{aligned}\mathcal{C}_t\left(\mathbf{s};\psi,\chi\right) = \quad & (1-\rho)\,\mathbb{E}_{\widehat{\delta}'|\widehat{\delta}}\left[EV_{t+1}\left(h+1, a-c_t, \widehat{\delta}'; \psi, \max\left\{\chi-1,0\right\}\right)\right] \\ & + \rho\mathbb{E}_{\psi',\widehat{\delta}'|\widehat{\delta}}\left[EV_{t+1}\left(0, a-c_t, \widehat{\delta}'; \psi', \max\left\{\chi-1,0\right\}\right)\right],\end{aligned} \tag{14}$$

where the user forms rational expectations over the future habit stock, outside value, and belief update. The next posterior mean $\widehat{\delta}'$ follows equation (10). Because users are naïve about temptation, they perceive themselves as one minute less tempted when approaching the next episode $t+1$, resulting in a perceived next-period temptation duration of $\max\{\chi-1,0\}$ in the continuation value (14). The value function associated with this unlocking stage is therefore:

$$W_t\left(\mathbf{s};\psi,\chi,\boldsymbol{\epsilon}\right) = \mathbb{1}\{a \geq c_t\}\max\left\{u_t(h,\widehat{\delta},\psi,\chi,\epsilon) + \mathcal{C}_t(h,a,\widehat{\delta};\psi,\chi), \epsilon_0\right\} + \mathbb{1}\{a < c_t\}\epsilon_0, \tag{15}$$

which, with $\epsilon \sim Gumbel(-\beta_\epsilon\gamma, \beta_\epsilon)$, yields the expected value function:

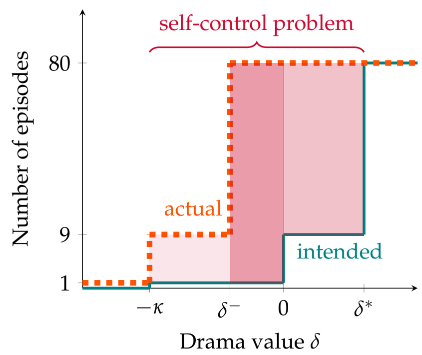
$$EW_t\left(\mathbf{s};\psi,\chi\right) = \mathbb{1}\left\{a \geq c_t\right\}\beta_\epsilon \log\left(1+\exp\left(\frac{\widehat{\delta}-\psi+\mathbb{1}\{\chi\geq 1\}\kappa+\alpha(h)+\mathcal{C}_t\left(\mathbf{s};\psi,\chi\right)}{\beta_\epsilon}\right)\right). \tag{16}$$

The solution of this single-agent dynamic model is characterized by the recursive system of equations (12)–(16), the belief transition (10), and the terminal condition (11). Numerically, one can solve these value functions and the corresponding policy rules by iterating over posterior mean $\widehat{\delta}$, temptation duration χ, and episode t, with posterior variance pinned down by t. I further characterize model solutions in Appendix D.1. There, I analyze users' top-up and unlocking choices, establish the cutoff values of posterior quality that govern package selection, and show how variation in temptation duration shapes demand for different package sizes. I also demonstrate that the unlocking decision follows a threshold rule in posterior mean, habit stock, and temptation.

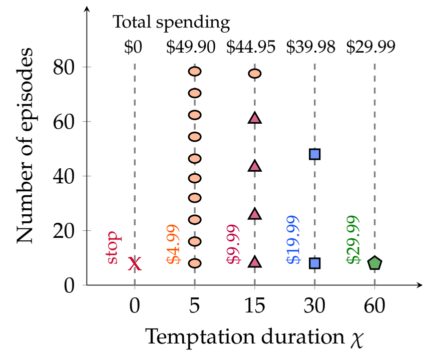
4.3 Predictions of Self-Control Problems

This section characterizes how self-control problems vary with drama value δ and temptation duration χ. These two dimensions generate distinct behavioral margins: overconsumption in the number of episodes watched and variation in top-up package choices. To isolate these mechanisms, I abstract from learning by imposing perfect information, set taste shocks ϵ and ν to zero, assume no initial token endowment, and fix each top-up package at the average token quantity.

Figure 6a illustrates an inverted-U relationship between drama value δ and the magnitude of the self-control problem, measured by the shaded gap between intended and actual episodes watched. Users with low valuations ($\delta < -\kappa$) do not continue beyond the first episode, even when tempted, and therefore do not suffer from a self-control problem. Users with high valuations



(a) Medium-valued users overconsume most



(b) Longer-tempted users overpay less

Figure 6: Illustrative examples: Self-control problems as functions of δ and χ

Notes: This figure illustrates self-control problems under different values of drama quality δ and temptation duration χ. To isolate the relevant mechanisms, I assume that drama quality δ is known to consumers, set all taste shocks (ϵ and ν) to zero, and assume no initial token endowment. Panel (a) plots the intended number of episodes watched (teal, solid line) and the actual number of episodes watched (orange, dashed line) for a tempted user with one-minute temptation ($\chi = 1$), as functions of δ. The shaded red area represents the magnitude of the self-control problem, measured as the gap between intended and actual consumption; darker shading indicates a more severe problem. Panel (b) illustrates the sequence of top-up choices for a user who, absent temptation, would watch only the free episodes but, under temptation, completes the entire series. The panel varies temptation duration χ and reports the corresponding total spending, showing that longer-lasting temptation is associated with lower overpayment.

($\delta > \delta^*$) rationally choose to watch all episodes, so temptation does not distort their behavior. Self-control problems instead arise primarily among users with intermediate valuations ($-\kappa < \delta < \delta^*$), for whom temptation drives a wedge between ex ante intentions and ex post consumption. In the example with one-minute temptation ($\chi = 1$), users with $-\kappa < \delta < 0$ intend to watch only the first episode, while users with $0 < \delta < \delta^*$ intend to watch only the free episodes, as shown by the teal line. Temptation shifts actual consumption upward, as shown by the orange line. Users with $-\kappa < \delta < \delta^-$ watch nine episodes rather than their intended one episode. Users with $\delta^- < \delta < \delta^*$ watch the entire series, despite intending to watch only one episode when $\delta^- < \delta < 0$ and only the free episodes when $0 < \delta < \delta^*$. This pattern has direct welfare implications: temptation-driven self-control problems are most severe for users with mid-range valuations. The structural estimates in Section 6.1 confirm this result.

Figure 6b shows how temptation duration χ affects top-up choices and overpayment. Without temptation ($\chi = 0$), the user watches only the free episodes. With temptation ($\chi > 0$), the user instead pays to finish the entire series, falling into the self-control trap. The anticipated number of episodes at the time of purchase varies with χ, so users with longer temptation spells are more likely to buy larger token packages. As a result, although all tempted users overconsume by completing the series, overpayment declines with χ: higher-χ users choose contracts with lower per-episode prices. This variation, generated by the menu of nonlinear contracts, helps identify the distribution of temptation duration and provides a novel contribution to the literature (e.g., DellaVigna and Malmendier, 2006).

Together, these examples show how drama value and temptation duration shape self-control problems through different behavioral margins. They also provide an identification argument. Early unlocking dynamics discipline learning, while repeated top-ups and persistent overpayment after substantial viewing, when users are already well informed, help distinguish temptation from residual uncertainty. I build the estimation strategy around this insight in Section 5.1.

4.4 Discussion

I discuss my model assumptions in this section, with the goals of being transparent in my modeling choices, reasonings, and potential biases.

Sophistication. I assume users are naïve about their temptation beyond its duration, due to insufficient variation in the data to identify the degree of sophistication. The observed overpayment behavior suggests that users are not fully sophisticated. The assumption of naïveté aligns with most findings in the behavioral literature (DellaVigna, 2018; Augenblick and Rabin, 2019) and is commonly imposed in structural analysis (DellaVigna, Lindner, Reizer, and Schmieder, 2017; Laibson, Chanwook Lee, Maxted, Repetto, and Tobacman, 2024). In Appendix D.5, I characterize the solutions for sophisticated users and estimate the structural model assuming that 20% of users are sophisticated. The results indicate that including sophisticated users implies stronger (11.4% higher) and longer-lasting (54.5% higher) temptation, because naïve users must exhibit greater temptation to generate the same level of self-control problems observed in the data. Incorporating sophistication therefore only reinforces my quantitative findings.

Naïveté about habit formation. I model irrationality in my framework through temptation and the associated naïveté. As summarized in Allcott, Gentzkow, and Song (2022), irrational digital addiction can also stem from naïveté about habit formation. Although this type of irrationality could also lead to repeated top-up purchases, we would expect users to gradually shift to higher-priced top-up plans as their habit stock grows over time. However, this pattern is inconsistent with what is shown in Figure 4d, where users frequently purchase another $4.99 package, regardless of how many episodes they have watched. Thus, I conclude naïve temptation is the more relevant story in the context of this short drama.

Round dynamics. I model round dynamics as exogenous, capturing users who pause and later resume viewing—potentially due to a temporary increase in the outside option. This assumption provides a tractable way to account for pausing behavior, which is not the focus of the analysis. It is not critical for the quantitative results, because all findings remain robust when rounds are defined using alternative thresholds, such as 12 hours. Alternatively, one could model pausing endogenously by assuming decreasing returns to viewing due to fatigue. My non-linear habit formation specification $\alpha(h)$ allows for this mechanism, whereas the estimation results in Section 5.2 reject decreasing returns.

Constant episode value. Instead of allowing the underlying drama quality to vary systematically by episode, I assume users learn about a constant latent value δ. This simplification is supported by the observation that viewership evolves smoothly across episodes in Figure 3a. If episode values differed systematically, we would expect greater volatility in viewership patterns. In the learning specification, η_t captures mean-zero episode-level experience noise used for belief updating, while the Gumbel shifter ϵ_t captures idiosyncratic taste shocks at the unlocking margin.

Network externality. Social media platforms often benefit from network externalities, which can contribute to user addiction (Barwick, Chen, Fu, and Li, 2024) or heighten the fear of "missing out" (Bursztyn, Handel, Jimenez, and Roth, 2023). I exclude this mechanism from my structural model for two main reasons. First, the platform analyzed here is small relative to mainstream social media, limiting potential network effects. Second, user utility on this platform is primarily driven by content–the short dramas–rather than social interaction, because users cannot message each other or leave comments on episodes.

Pacing. An alternative explanation for the repeated purchase of small top-up packages is that users are pacing themselves for reasons such as budgeting—that is, they lack sufficient cash on hand and therefore opt for lower-priced packages. However, this explanation is unlikely to be dominant in my context for two reasons. First, even the most expensive package ($29.99) is affordable for most adults. Second, as shown in Table 3, over half of users complete their viewership within a single round (a two-hour window), and more than 90% do so within two rounds. It is unlikely that users would engage in budgeting over such short time frames.

5 Estimation

I estimate my demand model using individual-level data from the short drama platform. The estimation approach is detailed in section 5.1, and the results are reported in section 5.2.

5.1 Estimation approach

To map my model to the data, I assume users, indexed by i, differ in their values δ_i, temptation durations χ_i, and initial endowments a_{i1}. For each individual, the data includes observations on their unlocking history $\boldsymbol{d}_i$, round $\boldsymbol{r}_i$, habit stock $\boldsymbol{h}_i$, token stock $\boldsymbol{a}_i$, and selected top-up packages $\boldsymbol{k}_i^*$ and token amounts $\boldsymbol{b}_{ik^*}$ for every episode t they have watched. I define my estimator and then discuss the identification strategy.

Estimator. The parameters to be estimated are:

$$\boldsymbol{\theta} := \{ \underbrace{\mu_\delta, \sigma_\delta, \sigma_\eta, \overline{\psi}, \beta_\epsilon}_{\text{intrinsic value and learning}}, \underbrace{\kappa, \alpha_\chi, \pi_0}_{\text{temptation}}, \underbrace{\alpha_1, \alpha_2, \rho}_{\text{habit}}, \underbrace{\omega}_{\text{price sensitivity}} \}.$$

For intrinsic value, the parameters μ_δ and σ_δ represent the mean and standard deviation of the latent drama value δ, which follows a Gaussian distribution $\mathcal{N}(\mu_\delta, \sigma_\delta^2)$. The learning signal has standard deviation σ_η, so each simulated individual has their own signal path and therefore their own posterior belief path $\widehat{\delta}_{it}$. The outside value during nighttime is captured by $\overline{\psi}$, and the dispersion of the unlocking taste shifter ϵ_t is captured by β_ϵ. Regarding temptation, κ represents its per-minute magnitude. Its duration has a point mass π_0 at zero and, conditional on being positive, follows a discrete Pareto distribution on $\{1, \ldots, 80\}$ with shape parameter α_χ, where all higher durations are absorbed at $\chi = 80$.[42] The habit-formation utility is modeled with a quadratic function parameterized by (α_1, α_2), and ρ controls for the likelihood that a round will end after watching an episode. Finally, ω represents price sensitivity. To normalize utility, I set the scale parameter of the top-up taste shifter, β_ν, to 1, so that it follows a standard Gumbel distribution.

Because the user-specific value δ_i is unobserved, the standard Maximum Likelihood Estimator (MLE) requires computationally intensive integration and is not well-suited to this setting. Therefore, I estimate my dynamic demand model using the Simulated Method of Moments (SMM) by minimizing a simulated criterion function:

$$\widehat{\boldsymbol{\theta}}_{SMM} = \arg\min_{\boldsymbol{\theta}} \left(\widehat{\boldsymbol{m}} - \widehat{\boldsymbol{m}}^S(\boldsymbol{\theta}) \right)' \boldsymbol{W} \left(\widehat{\boldsymbol{m}} - \widehat{\boldsymbol{m}}^S(\boldsymbol{\theta}) \right), \tag{17}$$

where $\widehat{\boldsymbol{m}}$ is a set of moments constructed from the observed data, and $\widehat{\boldsymbol{m}}^S(\boldsymbol{\theta})$ represents the average moments from S simulations of user behavior in the model (McFadden, 1989; Pakes and Pollard, 1989). The weighting matrix $\boldsymbol{W}$ is constructed via bootstrap, following Agarwal (2015). In each simulation, I simulate the sequence of top-up and unlocking decisions made by each user i, taking their initial endowment a_{i1} and the observed part of the round sequence from the data. The simulation also draws user-specific latent values, temptation durations, and episode-level learning signals, then updates $\widehat{\delta}_{it}$ recursively along each individual's simulated path. To ensure my estimator reaches a global solution to problem (17), I first conduct a series of grid searches to find an initial value close to the global optimizer, and then apply a local gradient-free Nelder-Mead algorithm to refine the optimization. Detailed estimation routines and intermediate results are documented in Appendix D.2.

I construct four categories of moments for estimation: habit stock, top-up package purchase, unlocking, and learning-related top-up dynamics. For habit stock, I target the probability of stopping after episode 9 among users with zero initial endowment, conditional on the user's last habit stock being $h_9 \leq 4$ or $h_9 \geq 5$. In addition, I include the average habit stock conditional on purchasing each top-up package. Second, for top-ups, I target the sales of each top-up package across different episode bins (10–20, 21–40, 41–60, and 61–80), which reveal the level and evolution of

[42] The quantitative results are robust with different ways of distributional parametrization.

top-up choices. Regarding unlocking, I track the share of users unlocking the second and third episodes, as well as episodes 4–9, 10–14, and 15–80. These episode ranges are intended to capture the most informative changes in unlocking behavior, as shown in Figure 3a. I also target the fraction of viewership that occurs during nighttime. Finally, to discipline learning and temptation, I add moments describing users' repeat and upgrade rates after the kth \$4.99 purchases as summarized in Figure 4e. A complete list of moments is provided in Appendix D.2. All parameters are jointly identified from the selected moments, though certain moments provide more information for specific parameters, for which I discuss the intuition below.

Intuition for identification. I first discuss how temptation is identified and why it can be separated from learning. In the model, temptation is the only source of time inconsistency and hence the only mechanism that generates self-control problems. The behavioral patterns documented in Section 3.5 therefore provide identifying variation for temptation. In particular, the positive market share of small top-up packages, together with repeated purchasing behavior, is informative about the magnitude of per-minute temptation, κ. Moreover, as discussed in Section 4.3, users with more persistent temptation are more likely to choose higher-priced packages. The relative shares of different packages therefore inform the distribution of temptation duration, α_χ. The mass at $\chi = 0$, π_0, is informed by the share of users whose choices are consistent with the absence of temptation. The key moment that separates temptation from learning is the patterns of repeated top-up purchases documented in Figure 4d and 4e, which can not be rationalized by learning alone as is shown in Section 5.2.

Habit formation (α_1, α_2), representing the channel of rational addiction, is identified through variation in habit stock. Empirical evidence suggests users with higher habit stock are more likely to continue watching, which makes the conditional probability of stopping for users at different habit stock levels informative.[43] Moreover, habit formation influences both the flow and continuation value, affecting users' top-up package purchasing decisions. As a result, the habit stock conditional on top-up choices provides information about (α_1, α_2) and the perceived probability of losing habit stock in the next period, ρ.

The learning parameter σ_η is disciplined by the dynamics of choices after users have observed early episodes and made their first small-package purchase. After receiving additional signals, users would update beliefs and systematically switch either to stopping or to larger packages. The repeat and upgrade moments after early \$4.99 purchases therefore help separate uncertainty about drama quality from persistent self-control problems. The intrinsic value is identified from the "residual" variation in unlocking and top-up choices after accounting for the temptation and habit formation. As discussed in section 4.2, users with higher δ are more likely to purchase the \$29.99 top-up package and unlock the entire series, whereas those with lower δ are more inclined to stop watching. Thus, the top-up shares of the \$29.99 package, the outside option, and the fraction of users who complete the series provide variation to identify this distribution $(\mu_\delta, \sigma_\delta)$. The nighttime outside value $\overline{\psi}$ is informed by the fraction of users unlocking episodes during this

[43]See Appendix C.2 for a complete discussion of related empirical patterns.

period. Lastly, β_ϵ captures the impact of the unlocking taste shifter on each episode, with more negative shocks increasing the likelihood that users will stop watching. As a result, the overall viewership dynamics offer information on the magnitude of this static utility shock.

Finally, the price sensitivity, ω, is reflected in users' top-up choices when they pay with real money, making the market shares of top-up plans at different prices informative. Because these packages are virtual products with no distinguishing characteristics beyond their observed prices and token amounts, the identification of ω is free from the usual concerns about endogeneity.[44]

5.2 Results

I report the estimation results for $\widehat{\boldsymbol{\theta}}_{SMM}$ in Table 5 and graphically in Figure 7. The targeted moments are displayed in Appendix D.3. The estimated price sensitivity is $\omega = 0.90$, which translates utility into real monetary terms (USD). I use this numéraire to report all quantitative results.

The first set of parameters describes the distribution of intrinsic value, which is also represented in Figure 7a. The estimated per-episode drama value δ has a mean of -0.020 (-2.2¢) and a standard deviation of 0.093 (10.3¢). The outside value during nighttime is 0.019 (2.1¢) per minute, whereas the daytime outside value is normalized to 0. The idiosyncratic taste shifter ϵ has a scale parameter of 0.046, resulting in a standard deviation of 0.059 (6.6¢). These results indicate that, on average, users experience a net loss of 2.8¢ per one-minute episode compared with their outside options, highlighting the potential loss in user surplus.[45]

The learning signal has a standard deviation of 0.039 (4.3¢), implying that users receive informative feedback about the latent value of a drama from each episode. This learning process is highly efficient: after the first episode, uncertainty about drama quality falls by 61%, and declines further to a negligible 1.3¢ by the time users make their first top-up choice. I therefore conclude that learning and uncertainty are not the main forces governing consumer behavior.

Temptation is estimated at 0.211 (23.3¢) per minute, placing it at the 99th percentile of the drama value distribution. As shown in Figure 7c, 28.8% of users have $\chi = 0$, while the remaining users are drawn from a heavy-tailed positive Pareto distribution capped at 80 minutes. The average duration is 11.2 minutes, with a standard deviation of 22.6 minutes. Approximately 22.9% of users experience temptation lasting more than 8 minutes (the duration covered by the \$4.99 package), 16.0% exceed 17 minutes (covered by the \$9.99 package), and 10.9% exceed 37 minutes (covered by the \$19.99 package). The cap at $\chi = 80$ absorbs 7.4% of users. The large magnitude and heterogeneous duration of temptation make self-control problems an important factor in drama-watching behaviors.

I also identify a substantial degree of habit formation, as shown in Figure 7d. The estimated

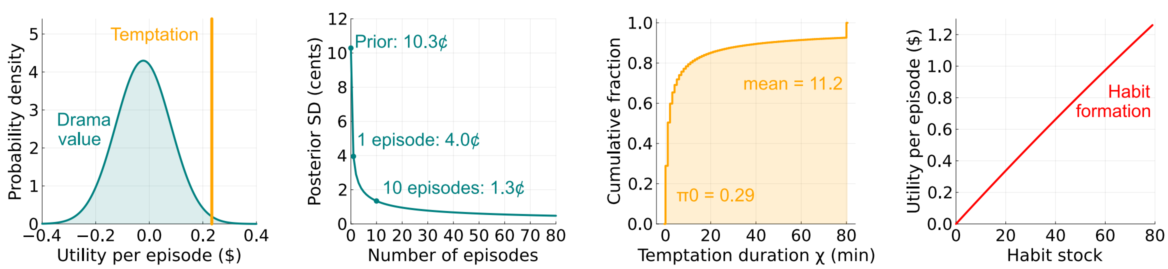
[44] In traditional markets (e.g., used cars), the identification of price sensitivity is subject to omitted variable bias: some attributes (e.g., car condition) are observed by buyers but unobserved by econometricians, and these attributes affect both choice and price. In contrast, top-up options in my setting differ only in two observable dimensions—price and number of tokens—eliminating concerns about such bias in identification.

[45] The average loss from watching one episode equals its average value of -2.2¢ minus the average value of the outside option, 0.27×2.1¢.

Table 5: Estimation: SMM estimators and standard error

Parameter	Estimator	Std	Meaning
1. Intrinsic value			
μ_δ	-0.020	0.010	Mean of latent drama value distribution
σ_δ	0.093	0.030	Standard deviation of latent drama value distribution
σ_η	0.039	0.0075	Standard deviation of episode-level learning signal
β_ϵ	0.046	0.0082	Magnitude of unlocking taste shifter
$\overline{\psi}$	0.019	0.057	Outside value during nighttime
2. Temptation			
κ	0.211	0.105	Magnitude of naive temptation
α_χ	0.516	0.107	Pareto shape of positive temptation duration
π_0	0.288	0.090	Mass of users with $\chi = 0$
3. Habit formation			
α_1	-1.40e-5	1.26e-6	Habit formation utility $\alpha(h) = \alpha_1 h (h - \alpha_2)$
α_2	1103.87	3.48e-5	Habit formation utility $\alpha(h) = \alpha_1 h (h - \alpha_2)$
ρ	0.046	0.0093	Probability of entering next round after watching
4. Price sensitivity			
ω	0.902	0.105	Price sensitivity

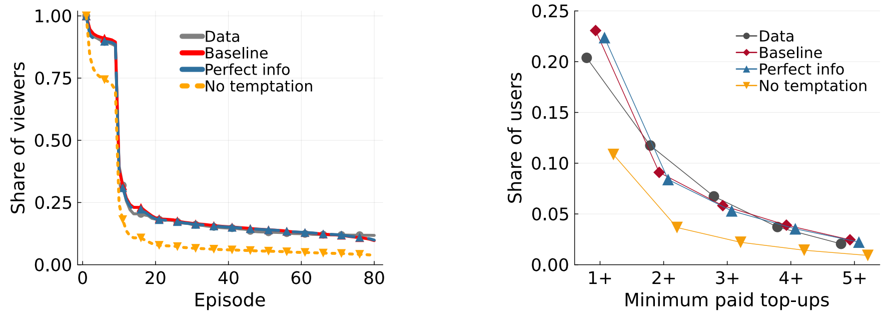
Notes: The GMM robust standard errors are reported in the table.



(a) Value (b) Learning (c) Temptation duration (d) Habit formation

Figure 7: Estimation: Intrinsic value, learning, temptation, and habit formation

Notes: Panel (a) shows the estimated distribution of latent drama value δ (teal curve) alongside the temptation level (orange bar), all normalized in dollars using the estimated price sensitivity. Panel (b) shows the decline in posterior uncertainty about δ as users observe episodes. Panel (c) displays the cumulative distribution of temptation duration χ, which has a mass at $\chi = 0$ and a positive Pareto tail capped at $\chi = 80$. Panel (d) illustrates the estimated function for habit formation $\alpha(h)$, which is nearly linear with respect to users' habit stock.



(a) Viewership

(b) Repeat top-ups

Figure 8: Temptation predicts persistence in viewership and top-ups

Notes: The figure compares observed data, the estimated baseline model, a perfect-information counterfactual that sets $\sigma_\eta = 0.0001$, and a no-temptation counterfactual that sets temptation duration to zero, holding the remaining estimated environment fixed. Panel (a) reports the share of viewers remaining by episode. Panel (b) reports the share of first-episode viewers with at least a given number of paid top-ups.

habit-formation function (depicted by the red curve) is convex and nearly linear in relation to habit stock, ranging from 0 to \$1.27 per episode. Watching an additional episode in the round contributes, on average, 1.6¢ in per-episode utility via habit formation. This strong level of habit formation aligns with the high willingness to pay observed among users in the data. The estimated probability of a round change, ρ, is 4.6%, influencing users' decisions through their expectations of future habit formation.

5.3 Temptation, not learning, predicts persistent behaviors

Figure 8 provides a decomposition exercise for the temptation mechanism by comparing the estimated model with two counterfactual environments. The first removes temptation while holding the rest of the estimated environment fixed. The second makes information perfect, while preserving the estimated temptation and habit-formation components. This comparison is useful because continuation and repeat top-up behavior could, in principle, be generated either by a short-run temptation force that induces persistence even when the rational surplus is low or by users gradually learning that a drama is valuable.

Panel (a) shows that removing temptation sharply reduces retention throughout the episode sequence, while the perfect-information economy remains close to the baseline. Panel (b) gives the same message on the payment margin. In the data, 20.4% of viewers make at least one paid top-up and 11.7% make at least two. The baseline model predicts 23.1% and 9.1%, respectively, whereas the no-temptation counterfactual predicts only 10.9% and 3.7%. By contrast, the perfect-information counterfactual remains close to the baseline, predicting 22.4% and 8.4%. Thus, the high persistence in paid top-ups is not primarily a consequence of slow learning about drama quality. Instead, it is disciplined by the estimated temptation component, which is needed to jointly match both continued viewing and repeated paid purchases.

6 Main results

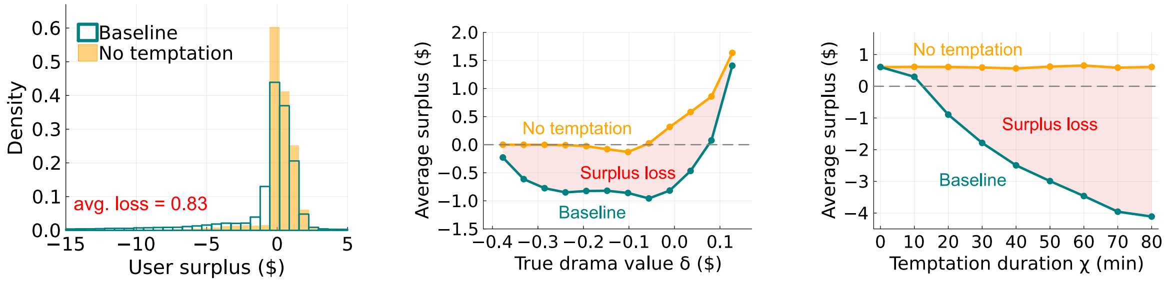
In this section, I present my quantitative findings on how short video length intensifies self-control problems. I begin by defining in section 6.1 the user surplus loss due to self-control problems. In section 6.2, I assess the comparative static over video length, demonstrating the magnitude of surplus loss due to the short form. Finally, in Section 6.3, I examine counterfactual policies aimed at mitigating self-control problems on the platform.

6.1 Surplus loss due to self-control problems

To evaluate welfare implications and efficiency loss due to self-control problems, I define user surplus as the utility from both intrinsic value and habit formation according to equation (8), net of token payments. This metric represents the surplus that a rational user would seek to maximize, treating temptation as an irrational component. I also net out the utility from the first episode, which is mechanically consumed in the model, so the welfare measure focuses on endogenous continuation and top-up decisions. My analysis is qualitatively robust with other measures of welfare, such as the pure surplus using only the intrinsic value.

Temptation can induce users to continue watching multiple episodes despite deriving negative intrinsic utility, leading to a potential loss in surplus. Figure 8a illustrates the share of remaining viewers per episode. The baseline model prediction (red line) closely replicates the observed data pattern (gray line), validating the effectiveness of my structural estimation. Removing temptation results in a substantial decline in viewer retention across episodes. For instance, without temptation, 16.4% of users stop watching after the first episode, versus 5.4% in the baseline. By the final episode, only 3.7% of users would complete the series, which is a 61.4% reduction from the baseline completion rate of 9.7%. This gap represents users who succumb to temptation and continue watching despite wishing they had stopped, indicating a significant surplus loss attributable to self-control problems.

I compute surplus loss using the same no-temptation counterfactual. As shown in Figure 9a, removing temptation shifts the surplus distribution to the right, reducing the mass of users with large negative surplus. The average user loses \$0.83 from temptation. Figures 9b and 9c show where these losses arise. By drama value, the no-temptation benchmark is close to zero for users with very low δ, because they stop after the free first episode once its utility is netted out. Temptation makes these users continue and pay, pushing baseline surplus below zero. The loss is largest for users near the margin of valuing the drama, for whom temptation induces substantial extra consumption that is not justified by their welfare-relevant utility, consistent with the inverted-U relationship discussed in Section 4.3. By temptation duration, the pattern is monotone: users with $\chi = 0$ have no surplus loss by construction, while the gap between the no-temptation benchmark and the baseline widens sharply for users with longer temptation duration. This pattern confirms the central mechanism of the model: temptation lowers welfare by sustaining engagement precisely for users whose rational continuation value is limited.



(a) Surplus distribution (b) By drama value (c) By duration

Figure 9: User surplus and the loss due to temptation

Notes: I compare the baseline model with a no-temptation counterfactual obtained by setting $\kappa = 0$, holding the estimated learning, habit-formation, and top-up environments fixed. User surplus excludes temptation and nets out the utility from the mechanically consumed first episode. Panel (a) plots the distribution of user surplus in the baseline model (teal outline) and in the no-temptation counterfactual (light orange). Panels (b) and (c) plot average surplus in the baseline model (teal) and in the no-temptation counterfactual (orange), respectively by true drama value δ and temptation duration χ. The light-red shaded area is the surplus loss due to temptation.

6.2 Short length amplifies self-control problems

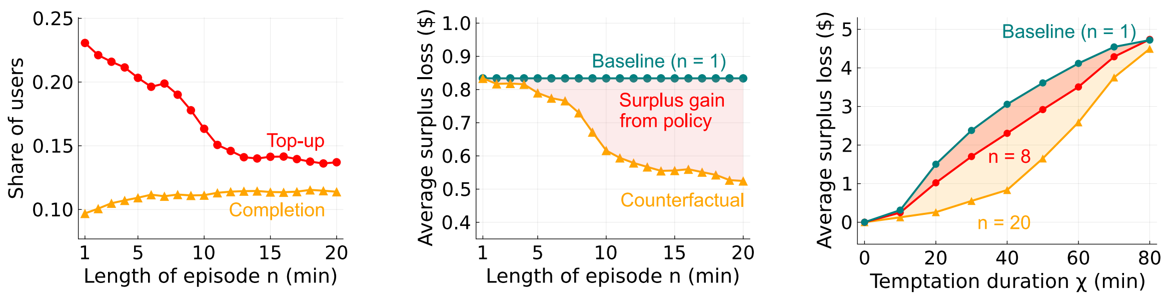
I now explore the mechanism by which short episode length amplifies self-control problems, using a comparative static exercise that varies episode duration. To avoid confounding effects from the platform's dynamic pricing rule, I hold the length of the first nine free episodes fixed. For subsequent episodes, let $n \in \mathbb{N}$ denote the counterfactual episode length, effectively merging n consecutive one-minute episodes into a single, longer episode. Let τ index these counterfactual episodes, with each τ corresponding to a set $Q_\tau \subset \{1, \ldots, T\}$ of original episodes. As in the stylized model in Section 2, I assume that once an episode is unlocked, the user watches it in full. The perceived flow utility is given by:

$$\widetilde{u}_\tau(h, \delta, \psi, \chi, \epsilon; n) = \mathbb{E}_\tau \left[n\delta - \sum_{t \in Q_\tau} \psi_t + \sum_{t \in Q_\tau} \alpha(h_t) + \min\{\chi, n\}\kappa + \sum_{t \in Q_\tau} \epsilon_t \right], \tag{18}$$

where the expectation is taken with respect to the information available in the first original episode of Q_τ, including habit stock, outside option, and taste shifter.[46] As in the stylized model, the key mechanism is a reduction in temptation due to the $\min\{\chi, n\}$ term, which dampens the effect of short-term bias. The counterfactual model is solved analogously to the baseline, using the same top-up framework.

Figure 10 reports the behavioral and welfare consequences of increasing episode length. Panel (a) first shows the behavioral margin. As n rises from 1 to 20 minutes, the share of users who make any top-up falls from 23.1% to 13.7%, while the completion rate rises modestly from 9.7% to 11.4%. Longer episodes therefore do not simply shut down consumption. Instead, they reduce

[46] I assume users observe the habit stock, outside value, and taste shifter associated with the first original episode in Q_τ and form beliefs about the rest, ensuring comparability between the counterfactual and baseline scenarios.



(a) User behavior (b) Average surplus loss (c) Loss by temptation duration

Figure 10: Comparative static: Episode length and self-control problems

Notes: These figures illustrate the counterfactual effects of setting a minimum episode length on user behavior and surplus. Panel (a) plots the share of users who top up and the share who complete the series. Panel (b) plots temptation-induced surplus loss, defined as no-temptation surplus minus baseline surplus at the same episode length. The teal line fixes the one-minute benchmark, the orange line shows the counterfactual loss, and the shaded area is the surplus gain from increasing episode length. Panel (c) plots the same surplus-loss object by temptation duration χ: the teal line is the $n = 1$ baseline, while the red and orange lines are the $n = 8$ and $n = 20$ counterfactuals. The red and orange shaded areas show the respective recovered losses and overlap where the gains are common.

the payment margin most associated with repeated temptation-driven continuation, while leaving completion approximately stable among users whose continuation value is sufficiently high.

Panel (b) translates this behavioral response into the welfare object used throughout the section: temptation-induced surplus loss, defined as no-temptation surplus minus full-model surplus at the same episode length. The one-minute benchmark implies an average loss of $0.83 per user, consistent with Section 6.1. Increasing episode length to eight minutes lowers this loss to $0.73, recovering $0.10 per user, or 12.4% of the benchmark loss. Extending the visual range to $n = 20$ makes the policy gradient more visible: the counterfactual loss falls to $0.52, recovering $0.31 relative to the one-minute benchmark.[47]

Panel (c) shows that the surplus gain from increasing episode length is hump-shaped in temptation duration. Very low-χ users have little surplus loss to recover, so the gain is small near $\chi = 0$. The gain then rises for lower- and intermediate-duration users: for these consumers, temptation is strong enough to distort continuation decisions in the one-minute baseline, but short enough that bundling episodes meaningfully reduces the frequency with which temptation affects choices. For very high-χ users, the gain declines because temptation remains active even within longer episodes, so increasing n only partially attenuates the self-control problem. This inverted-U pattern mirrors the mechanism in Proposition 2: short episodes are especially harmful when temptation is persistent relative to one-minute clips but not so persistent that longer episodes leave the temptation channel largely intact.

[47]For $n > 8$, the exercise also interacts with the top-up package menu, because the $4.99 package covers about seven one-minute episodes. I therefore use the $n = 8$ change as the main welfare comparison and interpret larger values of n as visualizing the policy gradient.

6.3 Counterfactual policy

I now turn to policy interventions designed to mitigate self-control problems. From the platform's perspective, I examine a minimum token package size requirement, which reduces self-control problems by increasing the episode bundle sold at each top-up. At the individual level, I consider a default time-limit policy, discussed in section 2.4, and a mandatory break following each top-up purchase, both intended to correct individual mistakes.

Minimum package size requirement. I first consider a class of policy that addresses the shortness of content by restricting the minimum size of token packages.[48] Because episodes are bundled and sold via these packages, raising the minimum package size effectively increases the duration of content per purchase, thereby mitigating self-control problems. I evaluate this policy by sequentially removing the lower-valued top-up packages ($4.99 and $9.99) in counterfactual simulations. Results are reported in section I of Table 6.

Column (2) reports the effect of removing the $4.99 package, which increases average user surplus by $0.46 (from −$0.25 to $0.21) and reduces average profit by $1.30 (from $5.32 to $4.01). As predicted by the model, eliminating the smallest package mitigates self-control problems, as reflected by a decline in the share of token purchasers from 23.1% to 14.2%. Those who do purchase tokens are selected on higher posterior continuation value and generally spend more, with average spending among token purchasers rising to $28.35. Column (3) further excludes the $9.99 package. User surplus increases to $0.32, driven by further reductions in self-control problems, while the share of token buyers falls to 12.2%.

Default time limit. Building on the screen time limit concept in Section 2.4, I consider a default time limit policy tied to the duration implied by the user's chosen top-up package. By default, users may watch only the free episodes and the number of episodes covered by their first top-up. Before viewing begins, they may opt out of the time limit by paying a cost valued at Φ dollars.[49] The decision problem for users who opt out is identical to the baseline problem in Section 4. I therefore focus on the solution for users who comply with the time limit, followed by their decision of whether to opt out.

In the model, this time limit is equivalent to restricting compliant users to a single top-up. I introduce an additional state variable $\ell_t \in \{0, 1\}$ to denote the number of remaining top-up opportunities. The state is therefore $(h_t, a_t, \widehat{\delta}_t, \ell_t)$, where $\widehat{\delta}_t$ evolves according to the same Bayesian rule as in equation (10). The system of value functions under this policy is defined below, with the

[48] Regulating product size is a practical policy tool. For instance, U.S. "loosies" law prohibits the sale of individual cigarettes and mandates a minimum pack size of 20 to discourage youth smoking and facilitate tax enforcement.

[49] This opt-out cost can be interpreted as a friction, such as a time delay, valued at Φ, or alternatively as a monetary fee charged by the platform to override the limit. Due to random taste shifters, adherence to this time limit requires a positive cost Φ.

Table 6: Policy evaluations: Minimum package size, default time limit, and mandatory break

Outcomes	Baseline	I. Min. Pack. Size		II. Time Limit		III. Break	
		$> \$4.99$	$> \$9.99$	$\Phi = 0.1$	$\Phi = 0.5$	1 min	5 min
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
I. *Per-user surplus*							
User surplus ($)	-0.25	0.21	0.32	-0.16	0.06	-0.20	-0.07
Profit ($)	5.32	4.01	3.64	4.54	3.87	5.09	4.61
II. *User activity*							
Top-up per token buyer ($)	23.05	28.35	29.73	22.47	21.09	23.13	23.14
Frac. token buyer	0.23	0.14	0.12	0.20	0.18	0.22	0.20
Frac. completion	0.10	0.10	0.10	0.10	0.10	0.09	0.09
Frac. time limit complier				0.77	0.99		

Notes: This table presents outcomes under three classes of policy interventions. Column (1) reports the baseline results without any policy. Columns (2) and (3) assess minimum package size restrictions by sequentially removing lower-priced top-up options, columns (4) and (5) evaluate default time limits with increasing opt-out costs Φ (in dollars), while columns (6) and (7) analyze mandatory breaks at top-up with different lengths. For each scenario, I report average user surplus, platform profit, average top-up spending per token purchaser, the share of users who purchase tokens, the share who complete the entire series, and, where applicable, the share who comply with the time limit.

superscript "TL" denoting the time limit policy:

$$\begin{aligned} V_t^{TL}(\mathbf{s},\ell;\psi,\chi,\boldsymbol{b},\boldsymbol{\nu}) = \quad & \mathbb{1}\{a<c_t\}\mathbb{1}\{\ell>0\} \max_{k\in\{0,\dots,K\}} \left\{EW_t^{TL}(h,a+b_k,\widehat{\delta},\ell-1;\psi,\chi)-\omega p_k+\nu_k\right\} \\ & + \mathbb{1}\{a\geq c_t\}EW_t^{TL}(h,a,\widehat{\delta},\ell;\psi,\chi), \end{aligned} \tag{19}$$

$$\begin{aligned} EV_t^{TL}(\mathbf{s},\ell;\psi,\chi) = \quad & \mathbb{1}\{a<c_t\}\mathbb{1}\{\ell>0\}\mathbb{E}_{\boldsymbol{b}}\left[\log\left(\sum_{k=0}^{K} e^{EW_t^{TL}(h,a+b_k,\widehat{\delta},\ell-1;\psi,\chi)-\omega p_k}\right)\right] \\ & + \mathbb{1}\{a\geq c_t\}EW_t^{TL}(h,a,\widehat{\delta},\ell;\psi,\chi), \end{aligned} \tag{20}$$

When $a < c_t$ and $\ell = 0$, the user can no longer top up, and the value of the outside option is normalized to zero. The functions $W_t^{TL}(\cdot)$ and $EW_t^{TL}(\cdot)$ are defined following equations (15) and (16), with $EV_t(\cdot)$ replaced by $EV_t^{TL}(\cdot)$ throughout and expectations taken over $\widehat{\delta}'$ conditional on the current posterior mean $\widehat{\delta}$. The key change is that users may top up only once, i.e., $\ell_1 = 1$. A user with initial endowment a_1, prior mean $\widehat{\delta}_1 = \mu_\delta$, temptation duration χ, and initial outside option ψ_1 will choose to opt out of the time limit if and only if:

$$\underbrace{EV_1^{TL}(0,a_1,\widehat{\delta}_1,1;\psi_1,\chi)}_{\text{Expected value with time limit}} < \underbrace{EV_1(0,a_1,\widehat{\delta}_1;\psi_1,\chi)}_{\text{Expected value without limit (baseline)}} - \underbrace{\omega\Phi}_{\text{Opt-out cost in util}}. \tag{21}$$

This condition typically applies to users whose posterior path makes repeated top-ups valuable. Users most affected by self-control problems, i.e., those who currently expect limited continuation value but repeatedly purchase small packages when temptation reappears, are most likely to comply with and benefit from the default time limit. Users whose beliefs remain high enough to justify completing the drama are less affected, as they tend to buy large packages and complete the series regardless of the restriction.

I report the counterfactual policy outcomes under varying opt-out costs in columns (4) and (5) of Table 6. Comparing column (4) with the baseline in column (1), even a small opt-out cost improves user surplus: a \$0.10 cost increases surplus by \$0.09 (from −\$0.25 to −\$0.16). However, the policy reduces per-user profit from \$5.32 to \$4.54, as it lowers top-up purchases on both the intensive and extensive margins. On the intensive margin, paying users reduce their average top-up amount from \$23.05 to \$22.47 due to mitigation of self-control problems. On the extensive margin, some compliers, who would have previously purchased the \$4.99 package anticipating no continuation value, now choose not to top up, further alleviating self-control problems. As the opt-out cost Φ increases to 0.5 in column (5), the fraction of compliers rises from 77.2% to 98.7%, further increasing user surplus to \$0.06 while reducing platform profits to \$3.87.

Mandatory break during top-ups. Finally, I consider a counterfactual policy that mandates a break following a top-up purchase by requiring users to watch an ad of ϕ minutes before resuming content. During this period, users experience their outside value but derive no intrinsic utility from the ad itself and do not feel tempted to continue watching it.[50] In this way, breaks serve as a mechanism to shorten the duration of temptation at the moment of the top-up decision.

To incorporate such a feature in my model, I adjust the top-up value functions in equations (12) and (13). The value of purchasing a package is evaluated at the shorter effective temptation duration $\max\{\chi - \phi, 0\}$:

$$
\begin{aligned}
V_t^B(\mathbf{s}; \psi, \chi, \boldsymbol{b}, \boldsymbol{\nu}) = \quad & \mathbb{1}\{a < c_t\} \max_{k \in \{0,\ldots,K\}} \left\{ EW_t(h, a + b_k, \widehat{\delta}; \psi, \max\{\chi - \phi, 0\}) - \omega p_k + \nu_k \right\} \\
& + \mathbb{1}\{a \geq c_t\} EW_t(h, a, \widehat{\delta}; \psi, \chi), \qquad (22) \\
EV_t^B(\mathbf{s}; \psi, \chi) = \quad & \mathbb{1}\{a < c_t\} \mathbb{E}_{\boldsymbol{b}} \left[\log \left(\sum_{k=0}^{K} e^{EW_t(h, a + b_k, \widehat{\delta}; \psi, \max\{\chi - \phi, 0\}) - \omega p_k} \right) \right] \\
& + \mathbb{1}\{a \geq c_t\} EW_t(h, a, \widehat{\delta}; \psi, \chi). \qquad (23)
\end{aligned}
$$

Combined with the unlocking value functions (15) and (16), the belief transition (10), and the terminal condition (11), these equations define the new system of value functions. The mandatory break therefore affects choices by reducing the perceived temptation duration into $\max\{\chi - \phi, 0\}$ when users consider a top-up. The solution to this system yields the decision rules for each individual user under this counterfactual policy.

The counterfactual outcomes are reported in columns (6) and (7) for varying break lengths. Relative to the no-break benchmark, column (6) shows a one-minute break increases user surplus from −\$0.25 to −\$0.20. The policy alleviates self-control problems by reducing the share of users who purchase tokens, as reflected in the decline in the fraction of token buyers from 23.1% to

[50] The assumption that users derive outside value during ads reflects the reality that they often engage in other activities while ads play, ensuring the policy does not mechanically alter welfare. This assumption likely underestimates the policy's effectiveness–if users actively dislike ads and experience negative utility, the policy's impact on reducing addiction would be even greater.

22.0%. Furthermore, column (7) suggests extending the break to five minutes raises user surplus to $-\$0.07$ and lowers the token-buyer share to 19.9%.

7 Extension: From short dramas to short videos

I extend my structural analysis on the short drama platform to a broader short video platform such as TikTok with following assumptions. First, I revert to the stylized model in section 2, omitting payment, habit formation, and learning from the structural framework.[51] Second, for the quantitative analysis, I use the estimated temptation parameter, κ, along with the distributions for intrinsic value ($x := \delta - \psi$) and temptation duration (χ) from my structural estimation reported in Table 5 and Figure 7. Because the stylized model is designed to isolate the content-length mechanism rather than platform entry and exit, I evaluate welfare for a one-hour short-video session, matching the average daily time spent on TikTok.[52]

In the one-hour short-video benchmark, average surplus is $-\$0.44$ per user. Approximately 19.2% of users choose not to watch. As shown in Figure 11a, eliminating temptation raises average hourly surplus to \$1.72, implying an average temptation-induced loss of \$2.16 per hour. The loss is concentrated: 41.6% of users experience a positive welfare loss, and among affected users the average loss is \$5.19 per hour, with the most affected users losing as much as \$14.01 per hour.

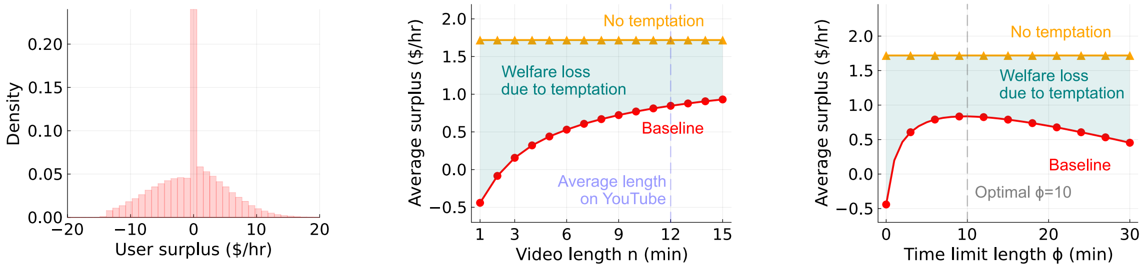
I begin by quantifying the effects of increasing video length, as outlined in Proposition 2. Figure 11b presents average hourly surplus for different video lengths, n. The baseline surplus curve rises with n, increasing from $-\$0.44$ at $n = 1$ to \$0.85 at $n = 12$, roughly the average length of YouTube videos.[53] In the no-temptation benchmark, average hourly surplus is \$1.72. Thus, increasing video length to 12 minutes reduces the hourly surplus loss due to temptation from \$2.16 to \$0.87, recovering \$1.29 per hour, or 59.6% of the one-minute temptation loss.

I next consider a practical policy: a default time limit on the short-video platform. As described in Section 2.4, the default ϕ-minute limit is modeled as a screen-time cap with the option to opt out. Proposition 3 shows that users with $x \in (-\kappa, 0)$ begin watching because temptation outweighs their intrinsic disutility. Users with temptation duration $\chi \leq \phi$ adhere to the default and stop after ϕ minutes, whereas users with $\chi > \phi$ opt out and continue for the full hour. The policy therefore reduces consumption among users with short temptation durations, mitigating self-control problems while preserving access for users whose high consumption is less affected by the default. Figure 11c evaluates this policy. The no-policy benchmark corresponds to $\phi = 0$, where all users with $\chi > 0$ opt out. Average surplus follows an inverted-U shape in the time limit: it initially rises as the default covers more users, but eventually declines because each complier consumes more short videos before stopping. The optimal default is 10 minutes, raising hourly

[51] The key distinction is that TikTok does not charge users, and TikTok videos are not interconnected like episodes in a drama series. Incorporating habit formation would amplify the platform's addictiveness, suggesting that the consumer surplus loss estimated here represents a lower bound.

[52] Source: https://www.emarketer.com/content/5-charts-on-video-marketing-momentum

[53] As of December 2018, the average video length on YouTube was 11.7 minutes (Source: Statista).



(a) Distribution of hourly surplus (b) Increasing video length n (c) Policy: Default time limit ϕ

Figure 11: Extension: Users' hourly surplus on short video platforms

Notes: Panel (a) shows the distribution of users' hourly surplus from short-video consumption, with a spike at \$0 reflecting the 19.2% of users who choose not to watch. Panels (b) and (c) report the counterfactual effects of increasing video length n and introducing a default time limit of length ϕ, respectively. In panels (b) and (c), the red curve shows average surplus in the baseline with temptation, the orange curve shows the no-temptation benchmark, and the shaded region represents the surplus loss due to temptation.

surplus from −\$0.44 to \$0.84. Equivalently, it reduces the hourly temptation-induced surplus loss from \$2.16 to \$0.88, recovering \$1.28 per hour, or 59.1% of the one-minute temptation loss.

Aggregate welfare implication. Finally, I examine the welfare implications for TikTok through a back-of-the-envelope calculation. According to a 2024 eMarketer survey, the average U.S. TikTok user spends 58.4 minutes per day, or 29.2 hours per month, on the platform.[54] Among users who watch in the model, this usage implies an average monthly surplus of −\$15.9 under the one-minute benchmark and a monthly surplus loss of \$78.0 due to temptation. Comparative statics suggest that short video length is a major driver of this loss. Increasing video length to 12 minutes raises monthly surplus to \$30.6 and recovers \$46.5 per user, or 59.6% of the monthly temptation loss. A 10-minute default time limit produces a similar gain, raising monthly surplus to \$30.2 and recovering \$46.2 per user, or 59.1% of the loss.

Given TikTok's large user base and the addictive nature of short videos, the aggregate welfare effects are substantial. With 150 million monthly active U.S. users reported in March 2023, the extrapolation implies a monthly temptation-induced welfare loss of \$11.7 billion. Imposing a 10-minute default time limit could recover \$6.9 billion per month, demonstrating significant potential to enhance user welfare. These calculations rely on strong extrapolation assumptions, but their magnitude indicates that policies targeting the interaction between short content length and temptation duration can have economically meaningful welfare effects.[55]

[54] Source: https://www.emarketer.com/content/5-charts-on-video-marketing-momentum

[55] The magnitude of the TikTok extrapolation is comparable to estimates for Facebook in Allcott, Braghieri, Eichmeyer, and Gentzkow (2020), providing a sanity check for this exercise.

8 Conclusion

This paper explores how small quantities amplify addiction, applying a theoretical framework to the short video industry. I find social media platforms offering short-form videos exploit users' self-control problems, leading to substantial losses in consumer surplus. Comparative statics reveal that shorter video lengths intensify user addiction, whereas policies mandating breaks can effectively reduce these self-control problems and alleviate the economic impact of social media addiction. My quantitative works provide an evaluation of potential policies, such as a default screen time limit, in reducing self-control problems. A follow-up experiment is desirable to verify the economic mechanisms and the magnitude of policy gain.

The implications of this study extend beyond social media. Evidence shows consumers frequently purchase small packages of goods (e.g., beer) at a premium, even when buying larger quantities would save money—a behavior likely driven by self-control problems similar to those observed in short video consumption. By connecting these diverse contexts, this paper highlights a broader economic principle: restrictions on quantity or access may serve as effective tools in curbing addictive behaviors across various industries.

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Appendix

Appendix A Model

Appendix A.1 Represent temptation with present-biased preferences

In this section, I demonstrate my model of temptation can be micro-founded by present bias within a class of standard quasi-hyperbolic discounting models. This formulation shows that the economic insights derived from my baseline model can be generalized within the classic present-biased preference framework.

Consider a large set of one-minute videos $\{1, ..., T\}$, where each minute represents a unit of time. At any given moment t, the agent chooses between watching videos and working (the outside option). Per-minute work provides delayed utility b after period $t + \chi + 1$, whereas watching a video yields immediate utility $b + x$. Consistent with the baseline model, x captures the intrinsic utility of watching a video relative to the agent's outside option.

Regarding time preferences, I assume the agent is present-biased, with the present defined as a window of χ minutes:

$$U_t = \sum_{\tau=t}^{t+\chi} u_\tau + \beta \sum_{\tau=t+\chi+1}^{T} u_\tau, \tag{A.1}$$

where the present-bias parameter $\beta < 1$ applies to all utility realized after χ minutes. This utility specification represents a quasi-hyperbolic discounting framework where the regular discount factor is set to 1 for clarity.

Given the preference structure in (A.1), the set of users experiencing self-control problems can be characterized as:

$$-(1 - \beta)b \leq x \leq 0. \tag{A.2}$$

Agents with $x \leq 0$ should not watch any videos. However, due to present bias, those with $x \geq -(1 - \beta)b$ find video-watching more appealing because $b + x \geq \beta b$, making them willing to watch videos for the next χ minutes. These users fall into a self-control trap, as their bias continually shifts toward the next set of future videos, leading them to consume more than intention.

This quasi-hyperbolic discounting specification serves as the micro foundation for my baseline model of temptation. The temptation duration χ can be equivalently interpreted as the span of present bias, whereas the magnitude of temptation κ is micro-founded in present bias, taking the value $(1 - \beta)b$. This equivalence demonstrates that the economic insights derived from my baseline model are directly applicable to the standard present-biased preference framework.

Appendix A.2 Solution for sophisticated users

Agents might not be fully naïve about their temptation. Let $\widetilde{\kappa} \in [0, \kappa]$ be the perceived temptation, where $\widetilde{\kappa} = \kappa$ corresponds to sophistication and $\widetilde{\kappa} \in (0, \kappa)$ corresponds to partial sophistication. The perceived flow utility (1) can be generalized into:

$$\widetilde{u}_\tau^t(x, \chi) = \underbrace{x}_{\text{intrinsic utility}} + \underbrace{\mathbb{1}\left\{\tau < t + \chi\right\}\widetilde{\kappa}}_{\text{perceived temptation}} + \underbrace{\mathbb{1}\left\{\tau < t + \chi\right\}(\kappa - \widetilde{\kappa})}_{\text{unperceived temptation}}. \tag{A.3}$$

For users who are susceptible to self-control problems ($x \in (-\kappa, 0)$) in Proposition 1, I consider the following two situations:

1. **Users with** $x \in (-\widetilde{\kappa}, 0)$ **decide not to watch.** Because temptation $\widetilde{\kappa}$ is perceived, these users anticipate that once they begin, the irrational temptation will persist for the next χ videos, preventing them from stopping. They thus evaluate their total utility as $\chi(x + \kappa) + (T - \chi)x < 0$ and choose not to watch.

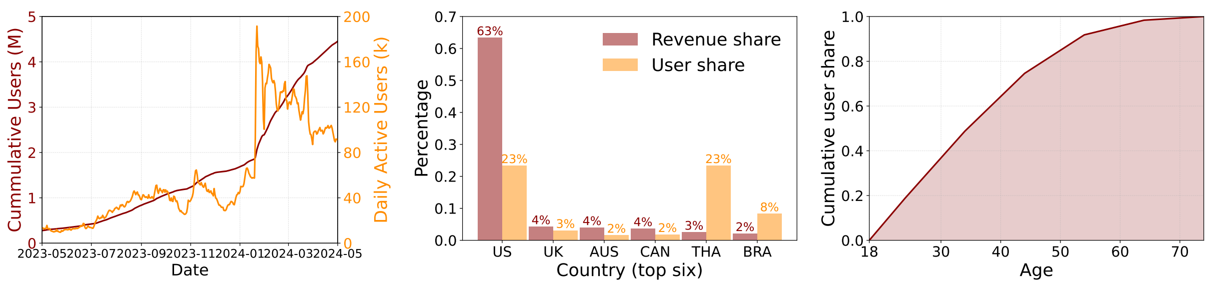
2. **Users with** $x \in (-\kappa, -\widetilde{\kappa})$ **fall into self-control problems.** Although these users recognize the temptation $\widetilde{\kappa}$, they do not expect to keep watching beyond the temptation duration χ, since the perceived temptation is insufficient to justify continuation. Yet the unperceived component is strong enough to trigger impulsive consumption, resulting in the self-control problem described in Proposition 1.

The general takeaway is that (partial) sophistication diminishes self-control problems. Under full sophistication ($\kappa = \widetilde{\kappa}$), users behave rationally and watch videos if and only if $x \geq 0$.

Appendix B Background

I provide further details on the platform in this section, including the descriptive facts on users, dramas, revenue, pricing, and the supply-side details.

Users. The platform size and user characteristics are summarized in Figure B.1. According to panel (a), the total number of users has reached 4 million worldwide and the Daily Active Users (DAU) remains around 100,000 by the May of 2024. Figure B.1b shows a large fraction of users are from US (23%), Thailand (23%), and Brazil (8%), whereas most of the platform revenue is contributed by US users (63%). Country boundaries create a natural form of market segmentation, and I therefore focus on the biggest market, the US market, in our main analysis. Panel B.1c indicates users of this platform are mainly young people, with 36% aged below 30, 63% below 40, and 85% below 50.

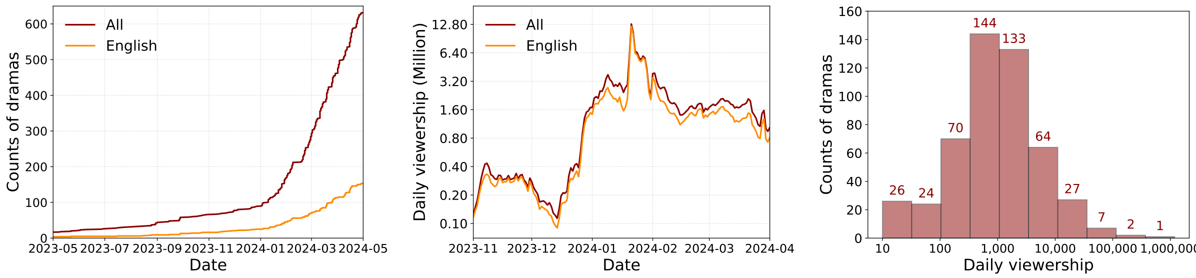


(a) Number of users (b) Geographical distribution (c) Cumulative distribution of age

Figure B.1: Users on the platform

Notes: Panel (a) shows the evolution of the accumulative number of users (left, red) and the daily active users (right, orange) between May 2023 and May 2024. My sample period is highlighted by shaded areas. Panel (b) reports the six countries with the highest revenues and most users up to May 8, 2024. Panel (c) displays the cumulative distribution function of age for new users who entered between March 1, 2024, and May 14, 2024. These aggregate data are directly provided by the platform.

Drama. I report the number and viewership of dramas in Figure B.2. Figure B.2a shows the evolution of the number of drama-language pairs over time. Over my sample period, the total number of drama-episode pairs increased from 67 in November 1, 2023, to 461 in March 31, 2024, where the number of English dramas increased enormously from 16 to 114. The corresponding viewership trends are displayed in Figure B.2b. Consistent with the geographical distribution in Figure B.1b, 83% of viewership during my sample period is in English. The daily viewership is 1.7 million episodes on average, which increased drastically in January 2024 due to the introduction of one "superstar" drama. I further report the distribution of daily viewership in Figure B.2c, which showcases a remarkable heterogeneity between dramas. The superstar drama, which was introduced on January 19, 2024, has about 1 million daily viewership, whereas most dramas have viewerships below 10,000 per day.



(a) Number of dramas (b) Aggregate viewership (c) Viewership distribution

Figure B.2: Drama entry and viewership

Notes: Panel (a) shows the evolution of the number of drama-language pairs on this platform over time. The shaded area indicates for the sample period for my study. Panel (b) reports the aggregate viewership of dramas over my sample period for all dramas and English dramas. Panel (c) plots the distribution of viewership on drama-language level. Note that viewership is displayed in log scale in both the y-axis of panel (b) and x-axis of panel (c).

Revenue. I present the time-series sequence of the three revenue sources in Figure B.3. The main source of all time is the token income, whereas the revenue from subscription has been increasing quickly since February 2024. The ad revenue remains at a relatively low level for the whole sample period.

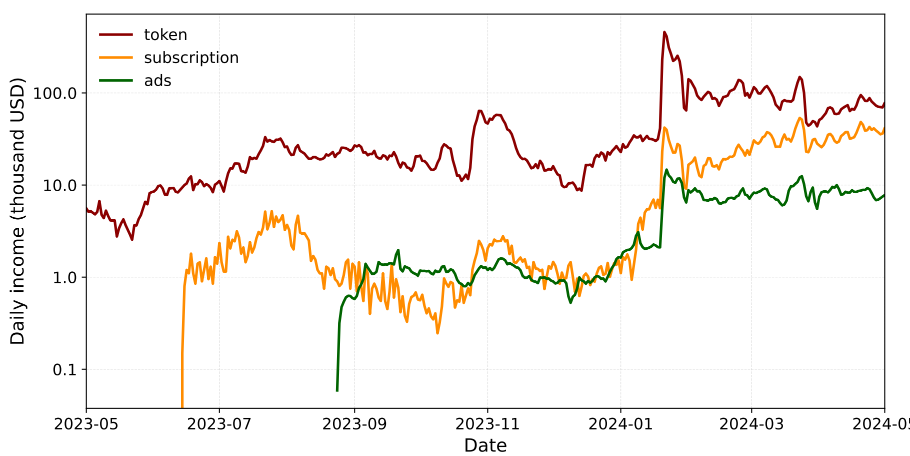
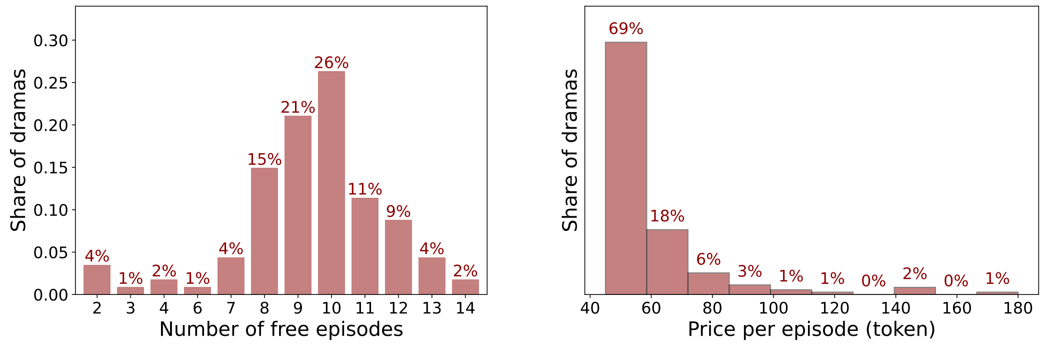


Figure B.3: Three sources of revenue: Token, subscription, and ads

Notes: This figure shows the evolution of different daily revenue sources. My data for subscription and ads started in June 2023 and August 2023, respectively. The level of revenue is displayed in log scale on the y-axis.

Drama pricing. In Figure B.4a, I report the length distribution of free episodes for each drama. Most dramas have eight to 10 free episodes to attract users, whereas the minimum and maximum numbers are two and 14. Figure B.4b displays the price distribution of episodes by dramas in terms of platform tokens. Within a drama, each episode costs the same amount of tokens to unlock, whereas price varies substantially across dramas. Most drama episodes are priced between 45 and 80 tokens.



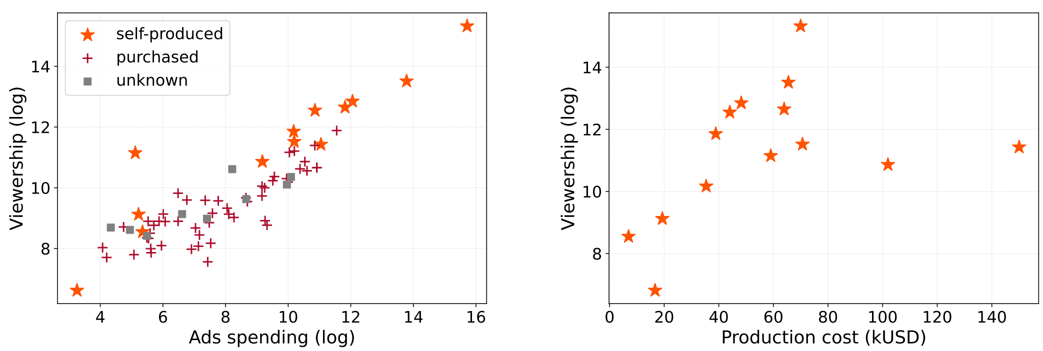
(a) Length of free episodes

(b) Price per episode

Figure B.4: Drama prices and token purchase

Notes: Panel (a) shows the distribution of the number of free episodes by dramas for all English dramas in my sample. Panel (b) plots the distribution of price per episode for those dramas.

Drama supply. In Figure B.5a, I show a positive relationship between ads spending and total viewership on drama level. Consistent with the aggregate description, self-produced dramas in general have higher ads spending and viewership. Conditional on ads spending, self-produced dramas also have higher viewerships than purchased ones, which indicates for some unobserved quality differences. I also observe the total production costs for platform-owned dramas, which are plotted in Figure B.5b against their total viewership. The production costs for dramas are fairly low, because most dramas are produced with costs below $80,000. In general, dramas that cost more are more popular among users, highlighting the need to consider heterogenous quality among different dramas.



(a) Ads spending

(b) Production cost

Figure B.5: Drama ownership, ads spending, and production cost

Notes: Panel (a) plots the total viewership and total ads spending between November 1, 2023, and March 31, 2024, for each drama. The plot distinguishes three types of ownership: self-produced (orange, star), purchased from independent producers (red, square), and dramas whose ownership is unknown to me (grey, circle). In panel (b), I plot the production costs for self-produced dramas against their total viewership.

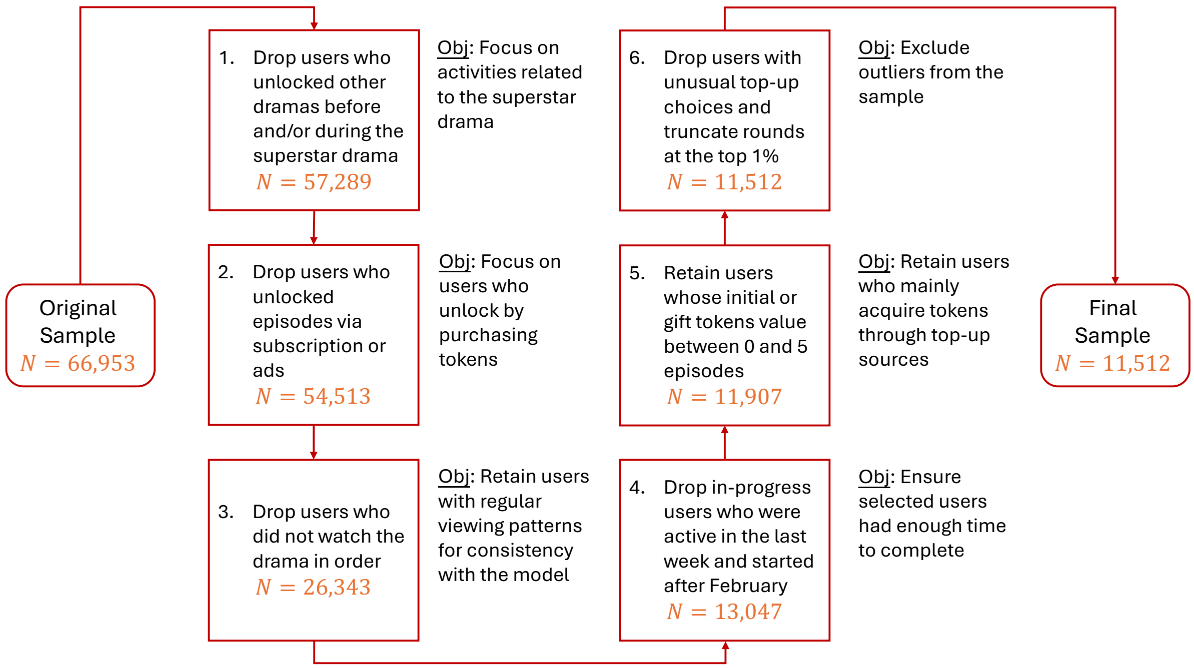


Figure C.1: Data cleaning procedure

Notes: This figure summarizes the data cleaning procedure step by step in the form of a diagram. Each box shows the main filtering step, the resulting sample size (in orange), and the objective of the step (in text alongside).

Appendix C Empirics

I present more empirical evidence from the platform to complements my main analysis.

Appendix C.1 Data cleaning procedure

This section provides a detailed discussion of the data cleaning procedure. The full process is summarized in six steps in Figure C.1, with each step accompanied by the resulting sample size and its underlying rationale. I elaborate on each step below.

1. *Drop users who unlocked other dramas before and/or during the superstar drama.* This step excludes users with prior experience watching other dramas on the platform, thereby removing potential confounds from pre-existing habit formation. It also ensures that all top-up activity is attributable solely to the superstar drama, aligning the sample with the structural model, which focuses exclusively on that title. The resulting sample remains broadly representative, retaining 85.6% of users, suggesting that this step is unlikely to introduce substantial selection bias.

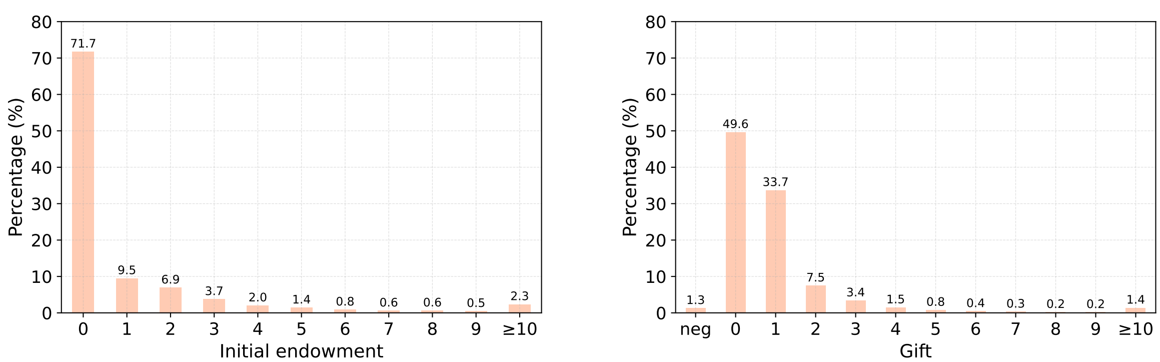
2. *Drop users who unlocked episodes via subscription or ads.* This step ensures that sample users unlock episodes exclusively through token purchases, which is the primary focus of the

analysis. As the vast majority of users (95.2%) use tokens only, this restriction is unlikely to introduce significant selection bias.

3. *Drop users who did not watch the drama in order.* This step retains users with regular, sequential viewing patterns, consistent with the structural model's assumption of episode-by-episode decision-making. It simplifies the modeling exercise by abstracting from the choice of viewing sequence. The cost is a substantial reduction in sample size, with 51.7% of users excluded. If users who watch out of order are less "rational" or more erratic, this selection likely results in a sample of more rational users—potentially leading to an underestimation of self-control problems, thereby reinforcing the paper's core argument.

4. *Drop in-progress users who were active in the last week (March 24–31, 2024) and those who started after February, 2024.* This step ensures that users in the sample had sufficient time to complete the drama, so that incomplete viewing reflects a choice not to continue. While this restriction reduces the sample size by 49.5%, it is unlikely to introduce systematic selection, because the timing of when users begin watching is plausibly random.

5. *Retain users whose initial or gift token value falls between 0 and 5 episodes.* This step excludes users who received a substantial number of tokens from non–top-up sources, such as daily log-ins, which appear as large initial endowments or gift token balances. Figure C.2 shows the distribution of these token values among users entering this step, indicating that most users receive minimal amounts from such sources. This restriction removes only 8.7% of users, suggesting that it is unlikely to introduce significant selection bias.



(a) Distribution of initial endowment

(b) Distribution of gift tokens

Figure C.2: Distribution of initial and gift tokens

Notes: Panels (a) and (b) report the distributions of initial endowments and gift tokens, respectively, among users entering Step 5 of the data cleaning procedure in Figure C.1. Token amounts are normalized by the number of tokens required to unlock one episode.

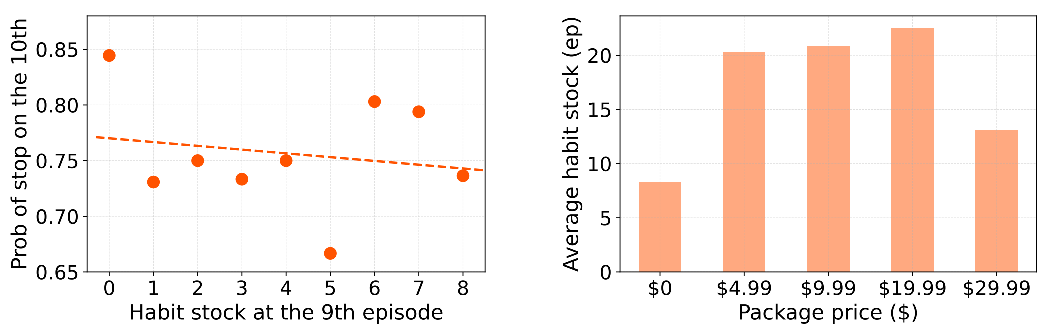
6. *Drop users with unusual top-up choices and truncate rounds at the top percentile.* This step removes outliers from the sample. Unusual top-up packages are defined as those with non-standard prices (i.e., outside the four main packages analyzed in this paper) or token amounts

with conditional frequencies below 5%. These packages typically originate from limited-time promotions or pricing experiments conducted by the platform. I also exclude users in the top percentile of the round distribution to eliminate extreme viewing patterns. This step introduces minimal selection, removing only 3.3% of users.

Appendix C.2 Rational addiction

In Figure C.3, I provide suggestive evidence of habit formation in users' drama-watching behaviors, which is commonly interpretted as rational addiction (Becker and Murphy, 1988). Panel C.3a shows the fits of regression described in section 3.5. Among users who have to top up before unlocking the 10^{th} episode of the superstar drama, a negative relationship exists between the likelihood of stopping and the corresponding habit stock. Those with one additional unit of habit stock are, on average, 0.3% more likely to continue watching.

Panel C.3b shows the average habit stock conditional on selecting each top-up package. A higher-priced package is in general selected by users with higher habit stock, supporting the habit-formation effect. The drop for the $29.99 package is mechanical, because users no longer need the largest package when they have already watched most episodes, and few remain to unlock.



(a) Conditional probability to stop

(b) Top-up package choice

Figure C.3: Suggestive evidence on rational addiction (habit formation)

Notes: Panel (a) shows that, among users who have to top up before unlocking the 10^{th} episode of the superstar drama, a negative relationship exists between the likelihood of stoping and the corresponding habit stock. Those with one additional unit of habit stock are, on average, 0.3% more likely to continue watching. Panel (b) presents the average habit stock based on the top-up package choice, where higher-priced packages are generally chosen by users with higher habit stock, indicating habit formation. The drop for the $29.99 package is mechanical, because users no longer need the largest package when they have already watched most episodes and few remain to unlock.

Appendix C.3 Viewership conditional on time of a day

Figure C.4 reports the number of unlockings conditional on hour of a day (Eastern time). It suggests the usage between 1am and 10am is below average, depending on which I define the "night-time" outside option for my structural analysis.

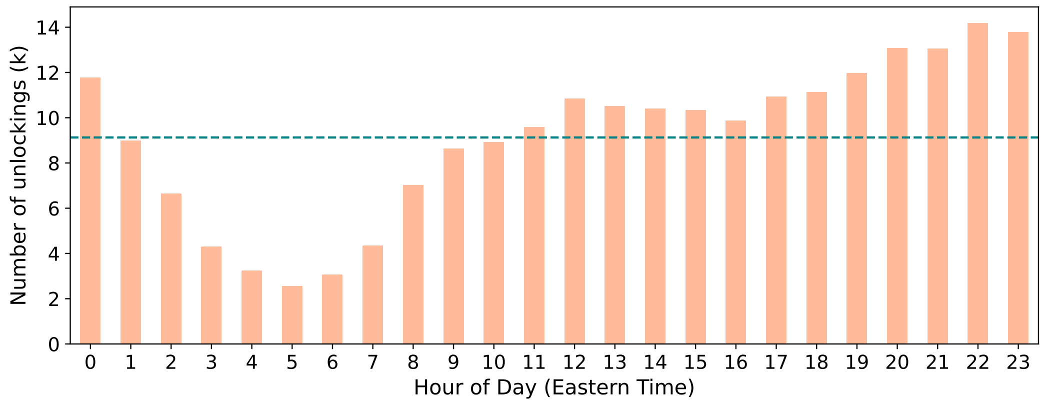


Figure C.4: Usage conditional on time of a day

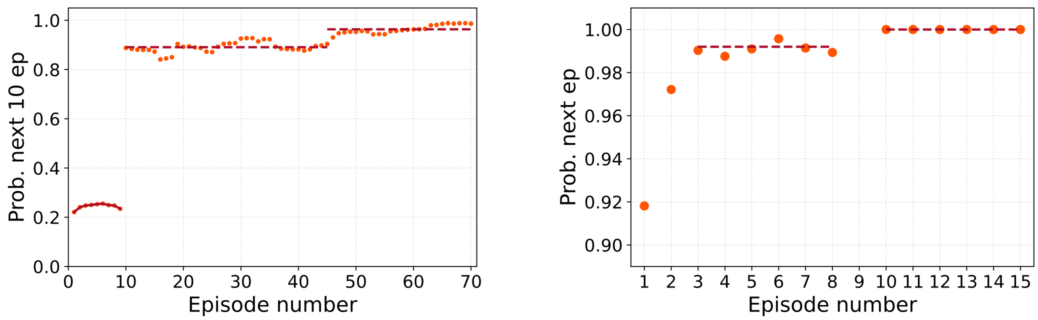
Notes: This figure represents the total number of unlocking activities that happened in each hour of a day within my sample. The hour is based on Eastern time. The teal, dashed line is the average number of unlocking per hour.

Appendix C.4 Uncertainty and Bayesian Learning

In this section, I provide evidence that learning or information effects are not the primary drivers of users' drama-watching behavior. Two key observations support this conclusion. First, the conditional probability of continuing to watch stabilizes within one or two episodes, indicating learning happens quickly, making the assumption of perfect information reasonable. Second, I provide conservative evidence for overpayment from an ex-ante perspective.

Evidence 1: The probability of continuing watching stabilizes quickly. In Figure C.5a, I present the evolution of the probability of continuing to the next 10 episodes, conditional on each episode. This probability initially rises from 22.1% at the first episode to 24.8% by the third, and then stabilizes around 25% until the ninth episode. A sharp increase occurs at the 10th episode—when users must start paying for new episodes—pushing the probability to 89.1%, where it remains stable between episodes 10 and 45. As the show nears its conclusion (episodes 45–70), the probability of continuing rises further to 96.3%. This analysis yields two key insights. First, the conditional probability increases significantly only during the first two episodes, after which it stabilizes quickly, suggesting learning occurs rapidly. Second, the price effect at the 10th episode is substantial, causing a 64.1-percentage-point increase in the continuation probability, whereas the learning effect accounts for only a few percentage points overall. Thus, I conclude learning plays a minimal role in shaping users' drama-watching behavior.

These insights are further confirmed in Figure C.5b, which illustrates the evolution of the conditional probability of continuing to the next episode. Initially, this probability is relatively low, at 91.8% for the first episode and 97.2% for the second. From the third episode onward, it stabilizes around 99%, indicating any learning effect occurs primarily within the first two episodes.



(a) Next 10 episodes (b) Next episode

Figure C.5: Conditional probability of watching the next few episodes

Notes: This set of figures presents the probability that a user will continue watching 10 (panel (a)) or one (panel (b)) conditional on watching each episode. The orange circles are the conditional probability, whereas the red, dashed lines provide fitted values. For a clean comparison, the conditional probability is calculated among users with zero initial token endowment.

Evidence 2: Ex-ante lower bound on overpayment. I next quantify a conservative lower bound on overpayment that allows for learning and other rational motives behind the smallest package. Following DellaVigna and Malmendier (2006), I ask whether users who purchased the $4.99 package would have spent less if their first $4.99 purchase in a given episode bin had instead been replaced by a larger package. This is an ex-ante exercise: it uses the package available at the purchase decision to infer whether the user could have made a cheaper commitment at that point.

Table C.1: Ex-ante lower bound for overpayment: Savings per user by replacing $4.99 package

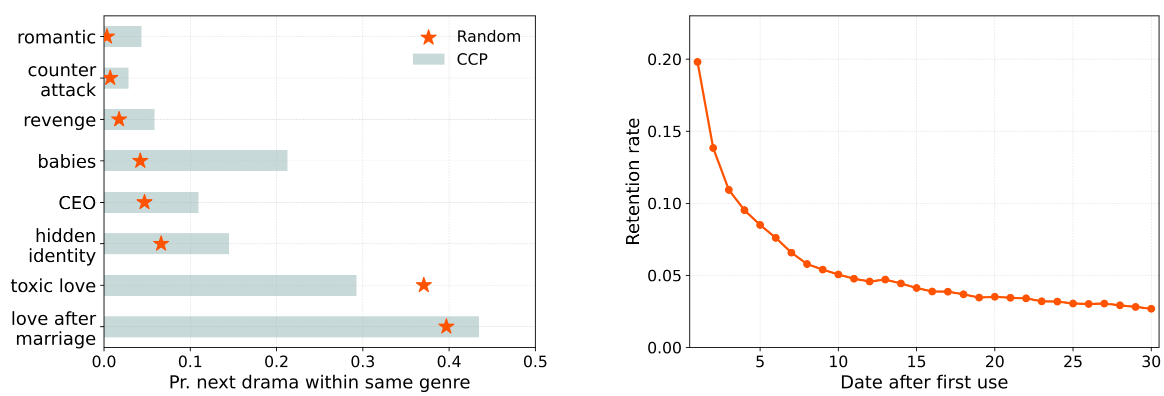
Episode (bin)		10-20	20-30	30-40	40-50	50-60	60-70	70-80
Replace $4.99 with:	$9.99	–1.72	–0.72	–0.84	0.20	0.30	2.62	–3.28
	$19.99	–6.71	–4.75	–4.04	1.75	–2.05	–6.16	–13.28
	$29.99	–10.05	–8.52	–8.70	–7.65	–12.05	–16.16	–23.28
Number of observations		797	203	140	97	74	83	154

Notes: In this exercise, I hold fixed each user's total number of episodes and replace their first $4.99 purchase within a given episode bin with a larger token package ($9.99, $19.99, or $29.99). All other top-ups are kept at their original choices before the switch or approximated by the closest original decision after the switch. To account for the discreteness of token purchases, I adjust unused tokens from additional packages by imputing their value at the user's average token price. The table reports average net savings per user, where negative values indicate higher costs. These estimates provide a lower bound on overpayment from self-control problems, as they abstract from uncertainty, learning, and other rational motives for choosing the $4.99 package. I highlight positive values in orange, indicating evidence of self-control problems even when alternative explanations may exist.

The key takeaway from Table C.1 is that the lower bound becomes positive for late $4.99 purchases. Replacing the first $4.99 package with the $9.99 package saves $0.20, $0.30, and $2.62 per user on average in the 40–50, 50–60, and 60–70 episode bins, respectively. The larger-package replacements are noisier because they impose stronger commitment, but the $19.99 replacement also generates positive average savings in the 40–50 bin. Negative values should not be read as evidence against self-control problems, because the exercise intentionally understates overpayment: users may buy the $4.99 package to learn about quality, may face uncertainty about outside options, and are not allowed to reoptimize after the counterfactual replacement. The positive late-bin entries are therefore conservative evidence that repeated small purchases generate ex-ante overpayment beyond what learning alone would predict.

Appendix C.5 No spillover across dramas

I do not find evidence for addiction beyond a drama, which justifies my analysis on the single, superstar drama. I check the history dependence on the level of both genre choice and platform usage. In Figure C.6a, I report the conditional choice probability (CCP) of a user who has just finished one drama to watch another drama that shares the same genre (the teal bar), along with the probability if the user just randomly selects the next drama following the empirical distribution (the orange star). The results show no sizable difference between the two, let alone the fact that user selection may mechanically make CCP greater than random probability. I therefore conclude no significant addiction occurs on the genre level. On the platform level, I plot the retention rates in Figure C.6b, which is defined as the fraction of users who are still active on the X^{th} day after their first appearance. The retention rate is fairly low relative to other digital platforms. For example, the 30-day retention rate on this platform is 2.7%, which is lower than Netflix (30.0%), Netflix (14.4%), and GoodNovel (7.5%). I therefore conclude no strong evidence exists for addiction beyond the drama level.

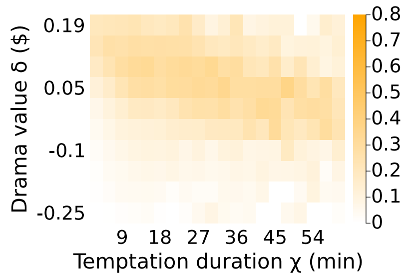


(a) No spillover within a genre

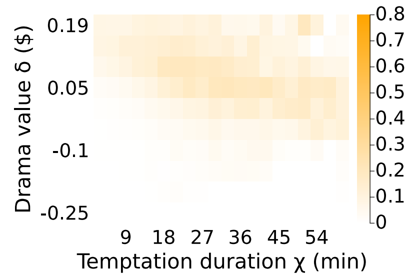
(b) No spillover within the platform

Figure C.6: No spillover across dramas

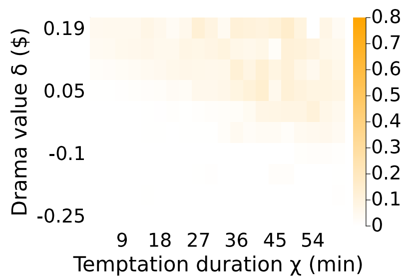
Notes: The teal bars in panel (a) show that, conditional on users just finishing a drama with the corresponding genre, the probability that the next drama they watch will share the same genre. The orange star indicates the probability if the users randomly select the next drama to watch based on the empirical distribution. Panel (b) plots the retention rate on the platform with varying number of days, which is the fraction of users who are still active X days after their first use. The 30-day retention rate is 2.7%, which is lower than other online platforms such as TikTok (30%), Netflix (14.4%), and GoodNovel (7.5%).



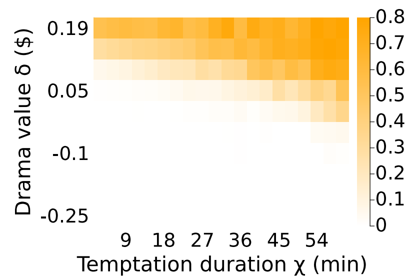
(a) Choice probability of $4.99



(b) Choice probability of $9.99



(c) Choice probability of $19.99



(d) Choice probability of $29.99

Figure D.7: Choice probability of each top-up package between episodes 10 and 15

Notes: The four heat maps report the estimated choice probability of each top-up package between episode 10 and 15 conditional on value δ and temptation duration χ. The warmer the color, the larger the choice probability. The probabilities from the four figures do not sum to 1, with the gap being the share that choose the outside option ($k = 0$).

Appendix D Structural Model

Appendix D.1 Solution characterization

Top-up package choices. With the full solution in hand, I can characterize the conditions under which users purchase each top-up package. For better intuition, consider the case without taste shifters ($\nu_k \equiv 0$). When $t > 9$ and the token balance is $a_t = 0 < c_t$, the perceived value of top-up package k is given by:

$$\pi_{kt}(h; \delta, \psi, \chi) = EW_t(h, b_k; \delta, \psi, \chi) - \omega p_k. \tag{D.1}$$

The top-up choice, therefore, depends on the shape of the value function $EW_t(h, b_k; \delta, \psi, \chi)$, which is monotonically increasing in all its arguments. For illustration, I plot the estimated choice probability for each package, conditional on user type, in Figure D.7.

The outside option ($k = 0$) yields a value of zero and will be selected if $\pi_k(h; \delta, \psi, \chi) < 0$ for all $k \in \{1, ..., K\}$, which typically occurs when users dislike the drama (low δ). When the intrinsic value δ is low enough, users will be unwilling to unlock the episode, regardless of their token stock, habit stock, or temptation duration. In this extreme scenario, $\lim_{\delta \to 0} \pi_{kt}(h; \delta, \psi, \chi) = -\omega p_k < 0$. Given the monotonicity and continuity, a cutoff value $\underline{\delta}(h, \psi, \chi)$ exists below which users will choose not to top up. As Figure D.7 shows, users with lower δ are more likely to select the outside option, as indicated by the lower choice probability for all packages.

At the other extreme, the largest package becomes most desirable when users plan to watch many episodes, which occurs when they derive high intrinsic value δ. When δ is sufficiently large, users believe ex ante they will finish the entire series, turning the problem into cost minimization. The highest-priced package, which offers the lowest price per token, is thus chosen early on, for example, around period $t = 10$. By monotonicity and continuity, a cutoff value $\overline{\delta}(h, \psi, \chi)$ must exist above which users opt for the largest package K. This intuition is supported by Figure D.7d, where most high-value users purchase the \$29.99 package between episode 10 and 15.

The medium-priced packages ($k \in \{1, \ldots, K-1\}$) are mostly purchased by users with medium intrinsic values, driven by varying durations of temptation. All else being equal, the longer this perceived temptation lasts, the more episodes the user anticipates watching. Mathematically, the value function $EW_t(h, a; \delta, \psi, \chi)$ increases more rapidly in a when $a \leq \chi$, and then slows down afterward. As a result, users with longer temptation durations are more likely to purchase higher-priced packages with more tokens. However, these users often continue topping up when running out of tokens again and ultimately finish the entire series, highlighting the self-control problem. This prediction is confirmed by the first three panels in Figure D.7, where the average temptation duration increases with the price of the top-up packages purchased by users. The choice probability of these medium-priced packages therefore informs the distribution of temptation duration.

Unlocking choices. With sufficient tokens, the user will unlock an episode if and only if their perceived value exceeds the outside option, namely, $u_t(h, \delta, \psi, \chi, \epsilon) + \mathcal{C}_t(h, a; \delta, \psi, \chi) > 0$. Because both the flow utility and continuation value are monotonically increasing in drama value δ, habit stock h, and temptation duration χ, the unlocking decision follows a threshold rule: users will continue unlocking if and only if the combination of intrinsic value, habit stock, and temptation is sufficiently high. For identification, the unlocking decision thus provides information on the relative magnitudes of habit formation, temptation, and the drama's intrinsic value.

Appendix D.2 Estimation algorithm

The targeted moment list below is updated to match the active learning specification. The two-stage search table in this subsection is retained as an estimation-path diagnostic and should be refreshed separately if it is used in the paper-facing appendix.

For estimation, I calculate model moments associated with each set of parameter $\boldsymbol{\theta}$ by simulating the economy 20 times and taking the average. To ensure my estimation routine converges to the global solution, I apply a two-step procedure that combines a global grid search with a subsequent local solver in approaching the SMM optimization problem (17).

Global grid search. I first conduct a global search over a large grid on my parameter space. I calculate the objective function on each grid point and locate the on-grid minimizer. The outcomeminimized objective function takes a value of 2361.51 and yields the initial estimators reported in Table D.2.

Local solver: Nelder-Mead algorithm. I then refine the first-step global minimizer $\widehat{\boldsymbol{\theta}}^{(1)}_{SMM}$ with a local solver. I apply a gradient-free solver, Nelder-Mead algorithm packed in `Optim.jl`, using programming software Julia. This process yields a final estimator with an objective value 2353.94.

Table D.2: Two-stage estimation procedure

	Parameter											
	μ_δ	σ_δ	$\overline{\psi}$	β_ϵ	κ	μ_χ	σ_χ	α_1	α_2	ρ	ω	obj.
First stage	-0.0094	0.091	0.017	0.056	0.15	1.21	1.26	0.000028	470	0.039	0.81	2361.51
Second stage	-0.0096	0.092	0.022	0.055	0.15	1.24	1.25	0.000028	472	0.039	0.81	2353.94

Appendix D.3 Targeted moments

Table D.3 reports the 40 targeted moments.

I report the targeted moments in Table D.3. I normalize the habit stock by the total number of episodes (80), top-up user share by the number of users in each episode bin, and unlocking share by the total number of users. The learning-specific rows report transition probabilities after the first and repeated \$4.99 purchases, together with all-user tail probabilities for repeated paid top-ups. Displayed values are rounded to two decimals.

Table D.3: Targeted moments

No	Moment	Note	Data	Model
1	`E_ccp_j10_h_leq_4`	Mean of stopping CCP for the 10th ep, conditional on last $h \leq 4$ and no endowment	0.76	0.80
2	`E_ccp_j10_h_5_8`	Mean of stopping CCP for the 10th ep, conditional on $5 \leq h \leq 8$ and no endowment	0.75	0.75
3	`E_h_k_0`	Mean of last habit stock when user chooses $k = 0$, normalized by T	0.09	0.10
4	`E_h_k_1`	Mean of last habit stock when user chooses $k = 1$	0.25	0.27
5	`E_h_k_2`	Mean of last habit stock when user chooses $k = 2$	0.27	0.28
6	`E_h_k_3`	Mean of last habit stock when user chooses $k = 3$	0.30	0.18
7	`E_h_k_4`	Mean of last habit stock when user chooses $k = 4$	0.17	0.10
8	`q_k1_j_10_20`	User share of package $k = 1$ sold at $t \in [10, 20]$	0.34	0.51
9	`q_k1_j_21_40`	User share of package $k = 1$ sold at $t \in [21, 40]$	0.18	0.30
10	`q_k1_j_41_60`	User share of package $k = 1$ sold at $t \in [41, 60]$	0.12	0.27
11	`q_k1_j_61_80`	User share of package $k = 1$ sold at $t \in [61, 80]$	0.19	0.26
12	`q_k2_j_10_20`	User share of package $k = 2$ sold at $t \in [10, 20]$	0.29	0.16
13	`q_k2_j_21_40`	User share of package $k = 2$ sold at $t \in [21, 40]$	0.23	0.14
14	`q_k2_j_41_60`	User share of package $k = 2$ sold at $t \in [41, 60]$	0.21	0.13
15	`q_k2_j_61_80`	User share of package $k = 2$ sold at $t \in [61, 80]$	0.17	0.06
16	`q_k3_j_10_20`	User share of package $k = 3$ sold at $t \in [10, 20]$	0.15	0.06

No	Moment	Note	Data	Model
17	`q_k3_j_21_40`	User share of package $k = 3$ sold at $t \in [21, 40]$	0.09	0.04
18	`q_k3_j_41_60`	User share of package $k = 3$ sold at $t \in [41, 60]$	0.21	0.01
19	`q_k3_j_61_80`	User share of package $k = 3$ sold at $t \in [61, 80]$	0.07	0.00
20	`q_k4_j_10_20`	User share of package $k = 4$ sold at $t \in [10, 20]$	0.15	0.32
21	`q_k4_j_21_40`	User share of package $k = 4$ sold at $t \in [21, 40]$	0.02	0.00
22	`q_k4_j_41_60`	User share of package $k = 4$ sold at $t \in [41, 60]$	0.04	0.00
23	`q_k4_j_61_80`	User share of package $k = 4$ sold at $t \in [61, 80]$	0.01	0.00
24	`unlock_2`	Share of unlockings at ep 2	0.94	0.95
25	`unlock_3`	Share of unlockings at ep 3	0.92	0.93
26	`unlock_4_9`	Share of unlockings at ep 4–9	0.89	0.91
27	`unlock_10_14`	Share of unlockings at ep 10–14	0.28	0.29
28	`unlock_15_80`	Share of unlockings at ep 15–80	0.14	0.15
29	`unlock_night`	Fraction of unlockings in the night	0.26	0.27
30	`first_k1_stop`	Next paid top-up outcome after first-ever $k = 1$: no next paid top-up	0.38	0.43
31	`first_k1_next_k1`	Next paid top-up outcome after first-ever $k = 1$: $k = 1$	0.30	0.34
32	`first_k1_next_k2`	Next paid top-up outcome after first-ever $k = 1$: $k = 2$	0.24	0.15
33	`first_k1_next_k3`	Next paid top-up outcome after first-ever $k = 1$: $k = 3$	0.06	0.04
34	`first_k1_next_k4`	Next paid top-up outcome after first-ever $k = 1$: $k = 4$	0.02	0.03
35	`repeat_k1_j12`	Next paid top-up is $k = 1$, conditional on current $k = 1$ and paid order $j \in \{1, 2\}$	0.32	0.36
36	`upgrade_k1_j12`	Next paid top-up is $k \in \{2, 3, 4\}$, conditional on current $k = 1$ and paid order $j \in \{1, 2\}$	0.32	0.24
37	`repeat_k1_j345`	Next paid top-up is $k = 1$, conditional on current $k = 1$ and paid order $j \in \{3, 4, 5\}$	0.45	0.47
38	`upgrade_k1_j345`	Next paid top-up is $k \in \{2, 3, 4\}$, conditional on current $k = 1$ and paid order $j \in \{3, 4, 5\}$	0.22	0.22
39	`repeat_topup_ge3_all_users`	All-user share with at least 3 paid top-ups; non-payers count as zero	0.07	0.06
40	`repeat_topup_ge5_all_users`	All-user share with at least 5 paid top-ups; non-payers count as zero	0.02	0.02

Appendix D.4 Decomposition of perceived utility

The estimated model suggests both rational and irrational addiction are significant in this short drama industry. I further decompose the perceived flow utility into three components: intrinsic value, temptation, and habit formation. Figure D.8 displays the average value of each component for users who have watched episode t. The teal area represents the average intrinsic value per episode, increasing from -1.2¢ for the first episode to 16¢ for the last, reflecting user selection over episodes as low-valuation users gradually stop watching. The sharp rise between episodes 10 and 14 is due to an exogenous increase in episode prices from 0 to 1 token, leading over three-quarters of users to stop watching. On average, users derive \$1.43 intrinsic utility from this drama relative to their outside options.

Addiction utility is substantial relative to intrinsic value. The orange area represents the value of temptation, valued at 19¢ per one-minute episode, providing an average perceived value of \$3.67 for each user from the drama series. However, this temptation is irrationally perceived and should be excluded from the subsequent welfare analysis. The red area at the top reflects

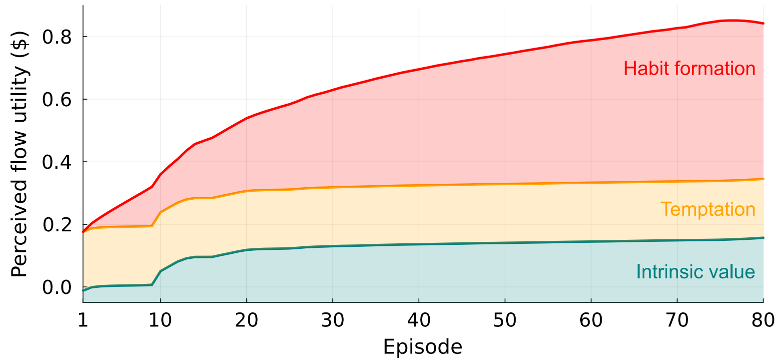


Figure D.8: Decomposition of perceived flow utility by episode

Notes: This figure decomposes the perceived flow utility into three components by episode: intrinsic value, temptation, and habit formation. For each episode, I report the average level of each utility component conditional on unlocking. All the components are normalized in dollars.

the average level of habit formation for each episode, which gradually increases as users build habit stock over episodes. For most episodes, habit formation is the primary component of utility, explaining users' high willingness to pay for this drama, conditional on paying. However, the majority of users stop watching between episodes 10 and 15, which reduces the average impact of habit formation across users. On average, users derive $4.30 from habit formation throughout the series, suggesting the realized value is similar between rational and irrational addiction.

Appendix D.5 Robustness: Sophistication

In this section, I discuss the structural estimation under sophistication, where users recognize that their future selves will face the same temptations when making unlocking and top-up decisions as current selves. I first present the model solution for sophisticated users. Although the data do not allow me to identify the degree of sophistication, I estimate the model assuming that 20% of users are sophisticated, showing that the naïveté assumption is conservative and provides a lower bound on the impacts of temptation.

Model solution for sophisticated users. I define the system of value functions recursively for sophisticated users, denoted by a superscript "$*$," building on the naïve solutions presented in section 4.2. Let χ^* and χ denote the actual and perceived temptation durations, respectively: the actual duration reflects the user's true susceptibility and shapes expectations about future behavior, while the perceived duration governs the utility calculation at time t. This distinction is crucial, as sophisticated users correctly anticipate χ^* when reasoning about future decisions but perceive temptation lasting only χ when evaluating utilities for current or future selves, which will be clear when we write the recursive system.

Following the same idea for naïve users, the (expected) value associated with the top-up decision writes:

$$V_t^*(h,a;\delta,\psi,\chi,\chi^*,\boldsymbol{b},\boldsymbol{\nu}) = \mathbb{1}\{a<c_t\}\max_{k\in\{0,1,\dots,K\}}\{EW_t^*(h,a+b_k;\delta,\psi,\chi,\chi^*)-\omega p_k+\nu_k\} + \mathbb{1}\{a\geq c_t\}EW_t^*(h,a;\delta,\psi,\chi,\chi^*), \tag{D.2}$$

and

$$EV_t^*(h,a;\delta,\psi,\chi,\chi^*) = \mathbb{1}\{a<c_t\}\mathbb{E}_{\boldsymbol{b}}\left[\log\left(\sum_{k=0}^{K}\exp\left[EW_t^*(h,a+b_k;\delta,\psi,\chi,\chi^*)-\omega p_k\right]\right)\right] + \mathbb{1}\{a\geq c_t\}EW_t^*(h,a;\delta,\psi,\chi,\chi^*). \tag{D.3}$$

The continuation value is adapted as:

$$\mathcal{C}_t^*(h,a;\delta,\psi,\chi,\chi^*) = (1-\rho)\,EV_{t+1}(h+1,a-c_t;\delta,\psi,\max\{\chi-1,0\},\chi^*) + \rho\mathbb{E}_{\psi'}\left[EV_{t+1}(0,a-c_t;\delta,\psi',\max\{\chi-1,0\},\chi^*)\right], \tag{D.4}$$

where the user perceives themselves with the actual length of temptation duration χ^* when making decisions in $t+1$, but calculate the utility with a shorter duration $\max\{\chi-1,0\}$ as the naïve user. Using this continuation value, we can define the value functions regarding unlocking decisions:

$$W_t^*(h,a;\delta,\psi,\chi,\chi^*,\boldsymbol{\epsilon}) = \mathbb{1}\{a\geq c_t\}\max\Big\{\underbrace{u_t(h,\delta,\psi,\chi,\epsilon)}_{\text{flow}}+\underbrace{\mathcal{C}_t^*(h,a;\delta,\psi,\chi,\chi^*)}_{\text{continuation}},\epsilon_0\Big\}\mathbb{1}\{a<c_t\}\epsilon_0, \tag{D.5}$$

and

$$EW_t^*(h,a;\delta,\psi,\chi,\chi^*) = \mathbb{1}\{a\geq c_t\}\beta_\epsilon\log\left(1+\exp\left(\frac{u_t(h,\delta,\psi,\chi,\epsilon)+\mathcal{C}_t^*(h,a;\delta,\psi,\chi,\chi^*)}{\beta_\epsilon}\right)\right), \tag{D.6}$$

where the flow utility depends on the perceived temptation duration χ.

Model estimation: When 20% of users are sophisticated. To assess how the naïveté assumption might bias the analysis, I estimate the model assuming that 20% of users are sophisticated. Solving the full system of sophisticated value functions (D.3), (D.4), and (D.6) is computationally challenging because the additional state variable greatly expands the state space. I therefore approximate sophisticated behavior by that of rational users with $\chi=0$. In Appendix A.2, I show that sophisticated users' behavior converges to rational behavior when T is large, as in early-stage decisions (e.g., the first top-up after free episodes). Moreover, sophisticated users purchase tokens only once, since they can correctly forecast future usage: they stop if they do not intend to continue, or purchase the largest package if they anticipate succumbing to temptation and finishing the series. This one-time decision makes subsequent top-ups less salient, supporting the suitability of the approximation.

Table D.4 reports the counterfactual estimation results. Assuming 20% of users are sophis-

Table D.4: Counterfactual estimation: When 20% of users are sophisticated

Parameter	Baseline		Sophistication		Meaning
	Estimator	Std	Estimator	Std	
1. Intrinsic					
μ_δ	-0.0097	0.0021	-0.0045	0.00045	Mean of drama value
σ_δ	0.092	0.0023	0.082	0.0017	SD of drama value
β_ϵ	0.055	0.0077	0.058	0.0062	Magnitude of unlocking taste shifter
$\overline{\psi}$	0.022	0.0045	0.016	0.00063	Outside value during nighttime
2. Temptation					
κ	0.15	0.0030	0.18	0.0041	Magnitude of naive temptation
μ_χ	1.24	0.022	1.69	0.030	Mean of temptation duration (log)
σ_χ	1.25	0.033	1.66	0.045	SD of temptation duration (log)
3. Habit					
α_1	-2.8e-5	4.3e-7	-3.0e-5	3.6e-7	Habit utility $\alpha(h) = \alpha_1 h (h - \alpha_2)$
α_2	471.94	2.9e-5	451.97	9.3e-6	Habit utility $\alpha(h) = \alpha_1 h (h - \alpha_2)$
ρ	0.039	0.00049	0.039	0.0011	Prob of entering next round
4. Price					
ω	0.81	0.0071	0.86	0.0033	Price sensitivity

Notes: The GMM robust standard errors are reported in the table.

ticated, most parameter estimates remain close to the baseline. The main difference lies in the temptation distribution, with per-minute temptation valued at 20.6¢ (11.4% higher) and an average duration of 10.2 minutes (54.5% higher). These findings suggest that the naïveté assumption yields conservative estimates of temptation and its duration, providing a lower bound on the associated efficiency loss.